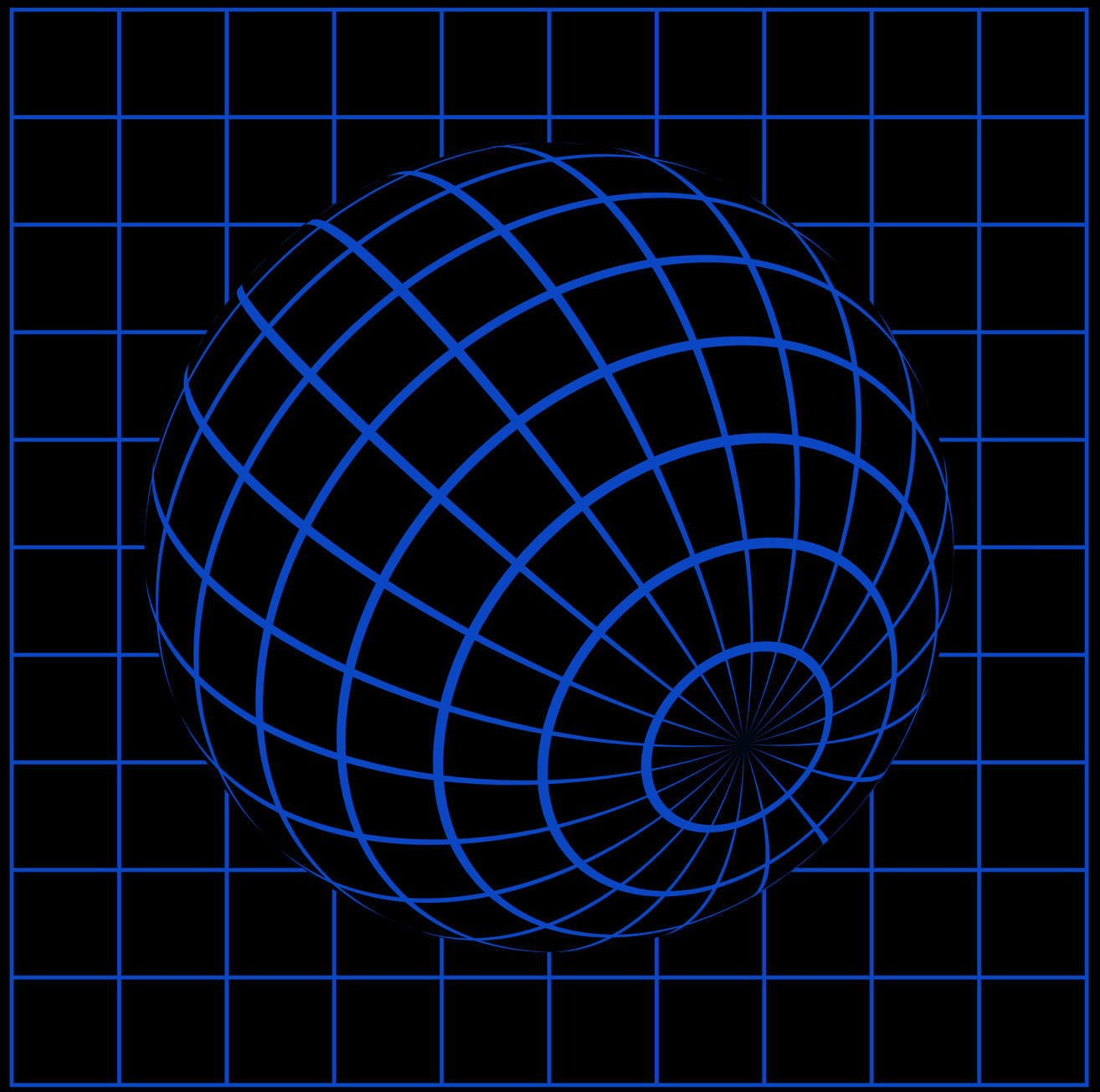
ROBLOCS



PORTFOLIO

shavRon®

Routed AI

TEAM IDENTITY

TEAM INTRO

At Roblocs, we aim to develop efficient planning, collaboration and management of tools along with sustainable and well-functioning engineering models, all of which aligns well with the team's ambition and thus help this team achieve its goals with ease.

KICK-OFF MEETING

All of the above, the scope, the main goal and the vision were discussed and analysed in the kick-off meeting. An online Google Meet was held with all the team members

TEAM MEMBERS



PROJECT CHARTER	
PROJECT NAME **TEAM NAME** **DATE**	National Robotics League Roblocs October, 1, 2025
TEAM MEMBERS	Arham Sethi - Autonomous Systems Engineer Aryav Shah - Mechanical Engineer Ayaana Soni - Robotics Engineer Agam Mundra - Robotics Engineer Siya Agrawal - Autonomous Systems Engineer Sahaj Agrawal - Brand Manager
PROJECT DESCRIPTION	Our project brings together adaptive software, rugged mechanical design, and real-time intelligence to create a robot that performs reliably under stress. We engineered a reinforced chassis, optimized motor geometry, and a high-efficiency gripper system for fast execution in every NRL challenge. The robot integrates sensor-driven autonomy, predictive control, and live telemetry to diagnose issues mid-run and adjust on the fly. Every subsystem, mobility, stability, sensing, and strategy, was redesigned from the ground up to survive impacts, recover quickly, and maintain precision.
PROJECT ROLE/JUSTIFICATION	Competing in the NRL, Team Roblocs aims to excel through advanced engineering, adaptive control systems, strategic design, and disciplined teamwork, building a battle-ready robot while developing real technical skills, innovation mindset, and competitive resilience.
DELIVERABLES	Deliverables I. Robot 2. Engineering Portfolio 3. Engineering Drawings 4. Code Files 5. Human-Robot interaction Mechnism 6. Digital Team logo 7. Social Media
ASSUMPTIONS	I. Sufficient Funding Availability 2. Timely Task Completion 3. Adequate Material Quality 4. Competitor Benchmarking Accuracy 5. Resource Availability 6. Immediate Mentor and coordinator Support
CONTRAINTS	The project is constrained by limited budget, time, resource availability, technical expertise, change in team members and roles, and strict compliance with competition guidelines.
RISKS	Risks include design and coding flaws affecting performance, budget overruns, delays in sensor arrival and testing, insufficient team coordination, technical challenges in meeting competition specifications, and inadequate quality of the manufactured parts

PRODUCT VISION

At Team Roblocs, we build competition robots engineered for resilience, precision, and adaptive intelligence. Our commitments:

Strength – Designing impact-resistant mechanisms, balanced weight distribution, and a chassis built to endure real arena chaos.

Control – Developing responsive drive systems, predictive sensing, and stable logic for consistent execution under pressure.

Reliability – Creating systems that self-diagnose, recover quickly, and maintain performance even after hits or failures.

OUR MISSION

Team Members			
Person	**Post**	**Skills**	**Job Function**
Arham Sethi	Autonomous Systems Engineer + Project Manager	Project management, coding, AI/ML, debugging, system coordination	Writes all autonomous-challenge code, maintains main robot codebase, handles debugging, integrates software with hardware, coordinates all departments, oversees innovation and execution
Siya Agrawal	Autonomous Systems Engineer	Coding, UI/UX, simulation, modelling, visualization	Builds creative software features, develops simulations, prototypes new coding concepts, supports debugging, implements new autonomous ideas, and tests algorithmic improvements
Aryav Shah	Mechanical Engineer	Material science, electrical engineering, electronics, hardware systems	Designs material layout, evaluates structural durability, manages electrical architecture, supervises mechanical build constraints, ensures reliability and compliance
Agam Mundra	Robotics Engineer	Robotics, mechatronics, material selection, CAD design, feasibility analysis	Leads innovation-challenge robot design, finalizes mechanical systems, performs feasibility checks, creates CAD models, defines mechanism architecture
Ayaana Soni	Robotics Engineer	Hardware testing, stress testing, assembly, electrical connections, CAD follow-ups	Performs robot assembly, conducts stress testing and hardware validation, maintains electrical connections, updates CAD-based adjustments, ensures battle-readiness of hardware
Sahaj Agrawal	Brand Manager	Cinematography, marketing, branding, sponsor acquisition, negotiation	Builds brand identity, creates visuals and videos, leads sponsor outreach, manages documentation quality, strengthens external communication and team presence

MEET THE ROBOT

1. Autonomous Code Assistance (Implemented + Prototype Elements Marked)

• Dynamic Forward/Backward Positioning (Implemented)

ToF “engagement bands” adjust creep-forward, hold, or slight retreat automatically during pickups.
• Fuzzy Soft-Approach Speed Profiling (Implemented)

Robot slows down smoothly as it nears the cube to prevent overshoot.

• Assistive Turn Dampening (Prototype)

Prototype algorithm to reduce micro-wobble and straighten approach paths in close-range control.

• Soft-Push Capture (Prototype)

Experimental low-speed forward nudge during gripping to auto-correct misalignment.

• Safety-Bounded Automation With Instant Override (Implemented)

All assist behaviours shut off instantly with joystick movement or Circle button.

2. One-Button Pickup & Stacking Automation (Implemented)

This automation removes guesswork and makes scoring actions consistent under pressure.

3. Autonomous Skill Challenge Superpowers

• Multi-Sensor Spur Authentication (Implemented)

Three-sensor fusion to confirm spur entry with very high reliability.

• Dynamic Speed Scaling Based on Line Error (Implemented)

Robot accelerates on stable straight segments and slows intelligently in turns.

• Adaptive PID Tier Selection (Implemented)

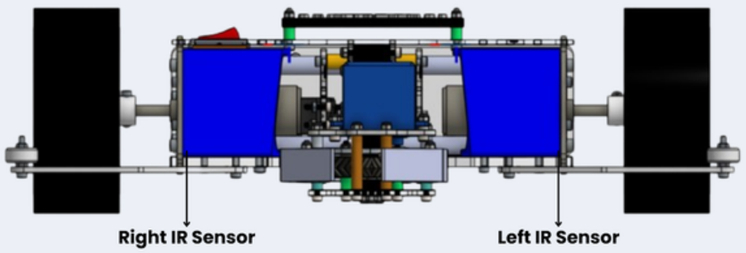
Controller automatically picks low, medium, or high authority gains based on instantaneous error.

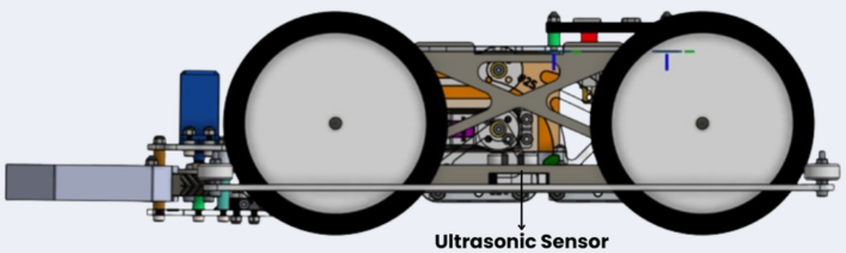
• Soft-Landing ToF Approach (Implemented)

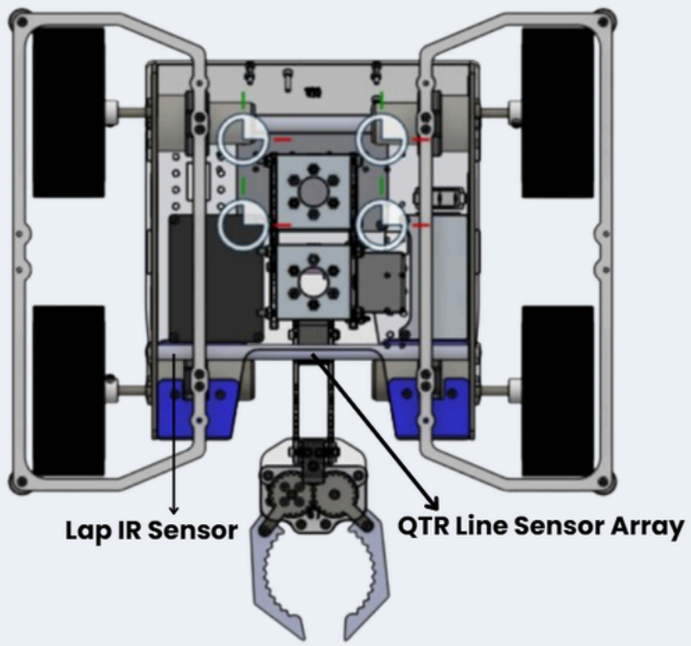
Smooth, decelerated approach into the charge zone.

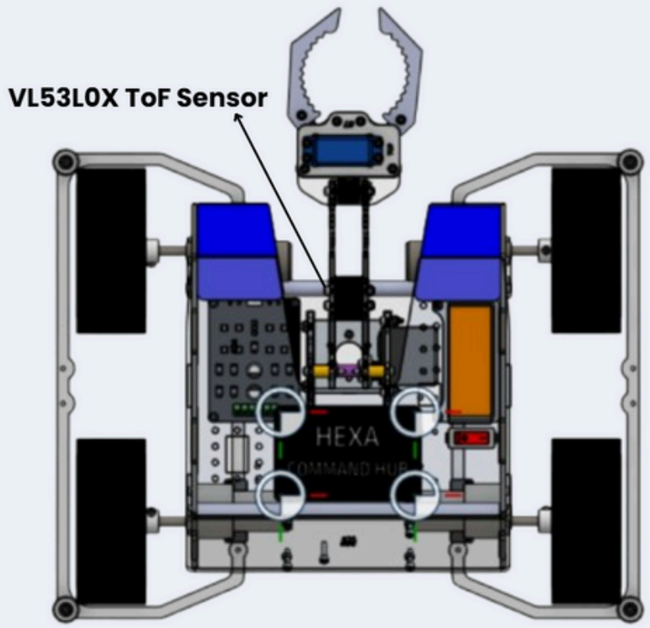
• Stability-Window Spur Exit & Line Rejoin (Partially Implemented)

Robot waits for a stable line signal before reaccelerating, preventing unstable rejoin events.







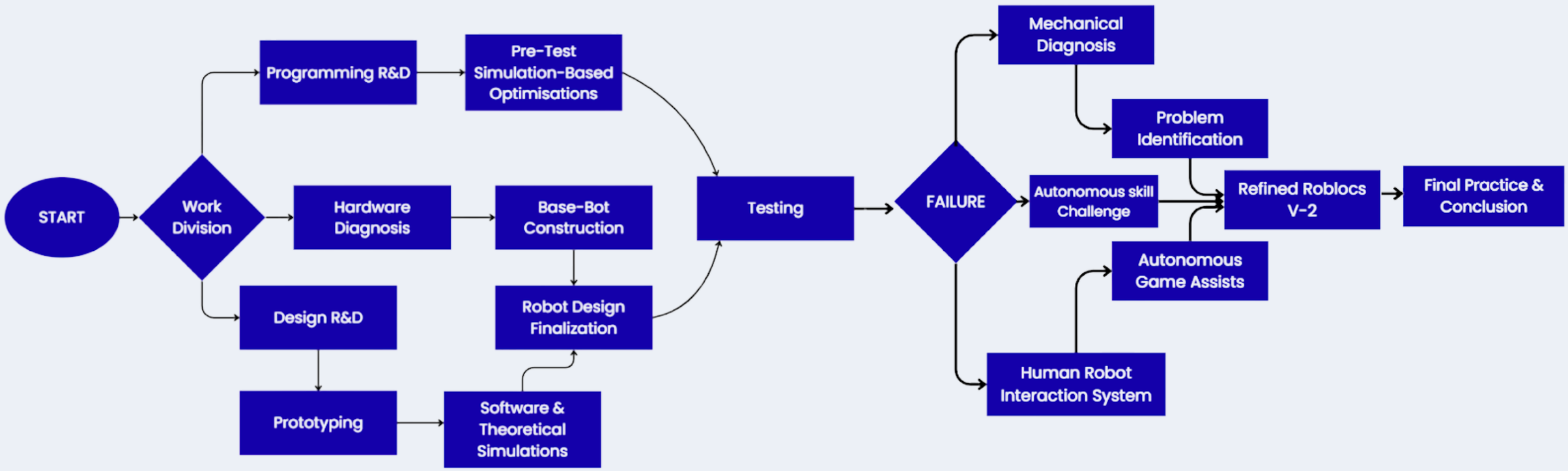


ENGINEERING STAGES

To present our engineering journey with the clarity it deserves, we structured the entire build process into well-defined stages. Each stage highlights the simultaneous development across Hardware, Design, and Programming, allowing us to capture not only what we built, but how and why we built it. This staged approach helped us analyse challenges as they emerged, document the reasoning behind every decision, and record the innovative methods we used to overcome obstacles.

By reflecting at the end of each stage, we identified key learnings that directly informed improvements in later phases. This chronological structure transforms our build log into a coherent narrative, one that shows the evolution of our robot from initial concepts to competition-ready performance. It ensures that anyone reading the document can follow our process step-by-step, understand the depth of engineering involved, and appreciate the continuous refinement that shaped the final BaseBot.

Timeline		
Phase	Start	End
1	18/10	1/11
2	1/11	12/11
3	12/11	19/11
4	19/11	24/11
5	24/11	29/11
6	28/11	5/11



STAGE 1

KIT ARRIVAL, RESEARCH & DEVELOPMENT

Marking the start of our engineering journey, the arrival of the kit also brought in high levels of enthusiasm, energy, and ambition. We took our time to understand the components, the given code, the working of the robot, and started our project on Notion.

Instead of just doing it the usual way, we tried to be innovative from the first day itself. When we sat together to understand the kit, the game, rules, and awards, we maintained a running list of all possible ideas and innovations in our first reading and made it the base for our Research & Development.

After getting over the kick-off phase, we divided ourselves into different departments for work efficiency. We had the hardware, design, programming, and marketing departments within our team, each with set deadlines, buffer periods, timelines, and objectives.

For the Research & Development, each innovation we conceptualized was scrutinized by 2 or more team members except the one who wrote it. The ideas were judged on potential value, time required, feasibility, and resources required. Based on these metrics, they were given a priority score, and fitted into the department and full team timeline.

HARDWARE

We compiled a document summarizing all of the technical regulations, and started doing technical analysis of major components provided in the kit.

1. BATTERY (LI-ION PACK)

Purpose in System
Primary energy source powering the drivetrain, servo actuation, motor driver, ESP32 control board, and peripheral electronics.
Engineering & Physics Evaluation
A stable Li-ion voltage profile ensures consistent current supply to the motors. Since DC motor torque obeys

$$\tau = k_t I,$$

and motor current depends on

$$I = \frac{V - E_b}{R_m},$$

any voltage sag directly reduces torque output. Li-ion cells exhibit low internal resistance, improving peak-load performance during acceleration and pushing interactions.

Reliability & Integration Considerations

- Secure fit in the chassis prevents vibration-induced disconnections.
- The guide advises teams to monitor charge cycles and keep multiple batteries ready for rapid rotation during events.
- External charger must be relied upon for BMS safety decisions, as the pack's internal protection is not user-visible.

2. SERVO MOTOR (#53)

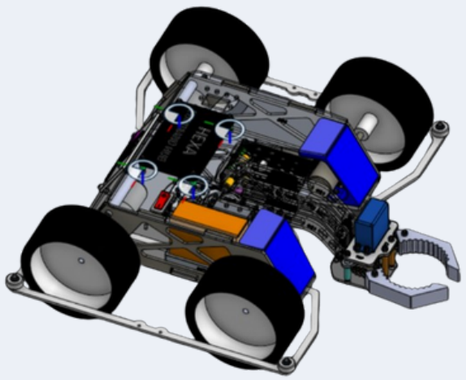
Purpose in System
Actuates the arm linkage and operates the gripper via helical gear reduction.
Engineering & Physics Evaluation
Lifting and manipulating a charge requires the servo torque to satisfy

$$\tau \geq W \cdot L,$$

where W is the load (≈100–150 g) and L is the arm's effective lever arm (≈10–15 cm). The stock servo meets minimum torque needs but remains close to its mechanical limits, especially during extended operation or off-axis loading.

Reliability & Integration Considerations

- Manual highlights avoiding manual servo manipulation, implying fragile internal gear composition.
- Helical gear pairs require consistent alignment; spacer compression or bolt loosening reduces reliability.
- Typical rotational speed (0.1–0.15 s per 60°) limits rapid-cycle stacking efficiency.



STAGE 1

3. ESP32 DEVKIT V1 (#48)

Purpose in System

Serves as the central controller for drivetrain actuation, servo control, telemetry acquisition, and wireless command processing.

Engineering & Computation Evaluation

The dual-core 240 MHz processor comfortably handles high-frequency tasks such as motor PWM generation, PID loops, PS5 controller input, and sensor fusion. Bluetooth latency (~3–4 ms) is negligible for competition driving.

The ESP32 provides over 30 usable GPIO pins, enabling integration of MDD10 motor driver, multiple servos, and any optional autonomous sensors such as ToF or reflectance arrays.

PWM signal quality is consistent across high frequencies, providing smooth motor control. Since average current follows

$$I_{\text{avg}} \propto D \cdot V,$$

where D is duty cycle, precise PWM control directly improves throttle resolution and manoeuvrability.

Reliability & Integration Considerations

- Must be shielded from metallic filings (construction guide explicitly recommends this).
- Requires a clean, regulated 5V rail to avoid brownout resets during high-load motor operation.
- Strongly compatible with the MDD10 motor driver, leading to predictable behaviour under dynamic load.

4. GRIPPER MECHANISM (POLYCARBONATE + HELICAL GEAR SYSTEM)

Purpose in System

Capture, secure, and release the charge elements during scoring cycles using a dual-finger assembly driven by helical gears.

Engineering & Physics Evaluation

The driving and follower helical gears provide smooth torque transfer with increased contact surface area. Helical meshes exhibit both radial and axial loads, making spacer stiffness and alignment critical.

The gripping force follows

$$F_{\text{grip}} = \frac{\tau_{\text{servo}} \cdot G}{r},$$

where G is the gear ratio and r the finger radius. While the mechanical advantage boosts grip strength, the limiting factor is servo output torque.

The flexible TPU fingers (#74) significantly improve energy absorption, surface conformity, and friction, benefiting high-vibration match conditions.

Reliability & Integration Considerations

- PLA gears may deform under high temperature or repeated stress cycles.
- Any misalignment of even 2–3° causes jamming or asymmetrical finger motion.
- Parallelogram linkage maintains consistent gripper orientation regardless of arm position

DESIGN:

Analysed the current robot design, and boiled down to the following limitations through the first analysis (without physically having the robot ready). We did an initial analysis even before we built the bot to ensure we immediately are able to deploy our ideas when the limitations show their effects. Our analysis hinted us to the following limitations:

$$\tau \geq W \cdot L,$$

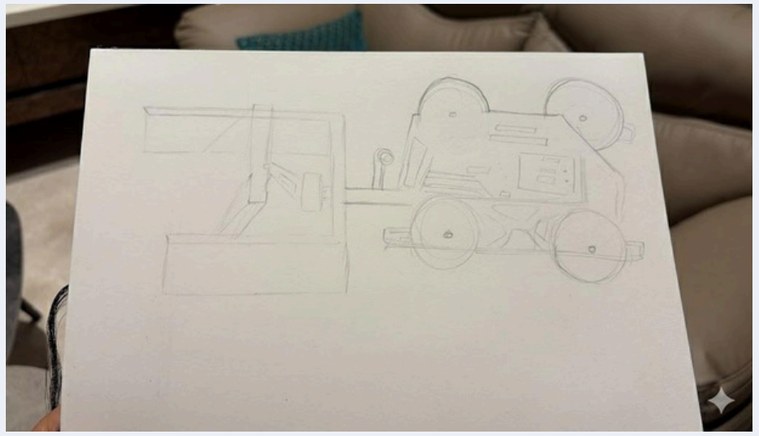
1. Only a single Li-ion pack is assumed; under aggressive driving and servo stalls, voltage sag can hit brownout thresholds for ESP32 and driver.
2. Motor driver and ESP32 are mounted openly on plates with high vibration and possible metal filings; guide explicitly tells you to cover the driver.
3. Each wheel is held by a single M3 bolt into the motor shaft; repeated impacts can loosen it, and there is no secondary keying or hub support.
4. Helical gears and fingers are PLA, driven by a single servo; gear tooth stress + axial loads + low friction on PLA fingers limit reliable grip force.
5. The entire arm + charge load hangs off one hobby servo and a 3D-printed front base block (#27) with long bolts; this concentrates bending stress and causes arm droop at extension.
6. Bottom/top plates are polycarbonate and motor holders are 3D-printed PLA, so under pushing loads the drivetrain can twist and the wheel axes can go slightly out of parallel.

STAGE 1

DESIGN EXTENSIONS TO BOT:

We explored three different mechanism concepts, one provided by NRL and two custom extensions we developed to improve efficiency, speed, and gameplay strategy.

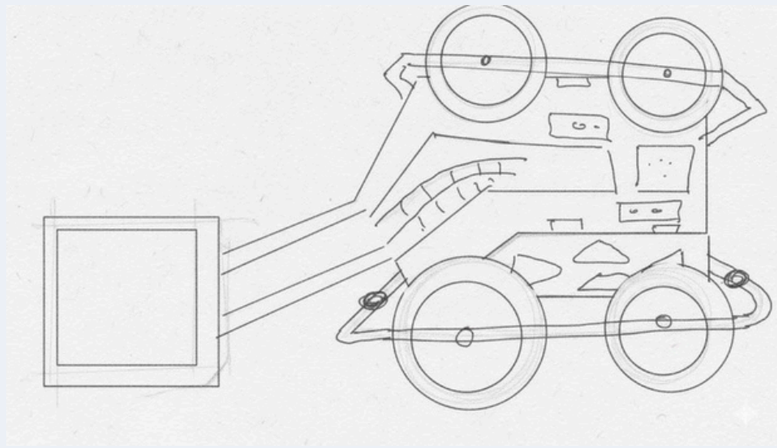
1. The first design is the basic claw that NRL gave to all teams. It's a simple grabber that picks up one cube and places it exactly where you want it. The good thing about this design is that it gives a lot of control, so the driver can be very accurate when scoring, especially for golden charges and stacking. Since it's the official mechanism, it naturally fits all the rules and works reliably as a starting point.



2. The second design which we thought of is a box that goes around the charge allowing the charge to be dragged across the gamefield. This one is meant to go over the charges quickly and push them across the field. The cubes will always be in front of the robot, which makes collecting and scoring much faster. It doesn't need extra servos or complicated movements, so it's fast, simple, and great for the early part of the match when the goal is to move charges quickly and control more of the field before the other teams can get to them.

3. The third design is the servo-powered launcher box. In this system, the robot doesn't have to drive all the way deep into the drop zone to place the cube. Instead, the cube sits inside the box, and when the robot reaches a certain distance from the scoring area, the servo extends a beam that launches the cube out of the box and fully into the drop zone. This means the robot can stop much earlier and still score, which saves a lot of driving time every cycle.

Because the servo does the final push, the driver doesn't need to perfectly align the robot or physically enter the corner every time. This makes the scoring process faster, more consistent, and easier for the driver. Compared to the claw, this system increases overall cycle speed since you spend less time maneuvering the robot and more time collecting the next charge.



PROGRAMMING:

In the first stage of programming, we thoroughly explored the starter code provided by the National Robotics League and broke down each module to understand how the drivetrain, servos, and controller inputs were structured. We customized the control mappings to match our drivers' instincts, ensuring smoother handling and faster reaction times during matches.

We also installed and verified all required libraries, configured the ESP32 environment, and tested serial communication. By the time mechanical assembly began, our entire software pipeline was already prepared, allowing us to upload, debug, and iterate the moment the robot powered on.

STAGE 2

CONSTRUCTION & BASEBOT ASSEMBLY

With our research & development groundwork complete, we began transforming concepts into a physical system by following the official construction sequence outlined in the NRL BaseBot manual. We transitioned from ideation to real engineering execution.

Although the instructions were clear, assembling a robot from 60+ mechanical parts, 100+ fasteners, toleranced 3D-printed components, and multiple actuators introduced real-world engineering challenges. Instead of treating the build as a 'follow-the-manual' task, we approached every step like a stage-by-stage system inspection. We checked tolerances, reinforced weak points, and ensured reliability from the very first prototype.

We maintained our departmental structure from Stage 1, however, hardware teams collaborated dynamically across subsystems to ensure workflow speed and consistency.

HARDWARE:

1. GRIPPER ASSEMBLY

We started with the gripper because it is the most tolerance-sensitive part. The helical gear meshing, connector link symmetry, and servo-horn alignment all require precision. Using parts #1 to #12 and #53, we assembled the dual-finger mechanism, ensuring:

- Proper alignment of helical gears (#4, #5)
- Smooth rotation along spacers (#11, #12, #13)
- Balanced torque distribution on the servo horn (#10)

Problems Faced:

Helical Gear Backlash & Stiffness

Because the printed gears are PLA, small imperfections caused partial binding during initial rotation.

Our Innovative Fix:

We placed a thin sheet of paper between the teeth during temporary assembly to automatically space the gears. When we removed it, it left a perfect micro-clearance.

2. ARM LINK ASSEMBLY

Using arm links (#21), parallelogram plates (#9), and spacers (#22, #23, #29), we assembled the four-bar linkage. The challenge was maintaining a proper alignment so that the gripper remains level throughout its motion arc.

PHOTOS

STAGE 2

3. CHASSIS ASSEMBLY

We then constructed the chassis using the bottom plate (#36), side plates (#42), back plate (#37), and base blocks (#27). The key task here was ensuring the correct alignment of the drive side plates so everything remained symmetric. Also, we had to ensure that all the screws and nuts were placed correctly with the components.
Problems Faced:
Nyloc Nuts Not Gripping Properly
We noticed something confusing at first. Even after tightening the M3/M4 nyloc nuts fully, a part of the screw thread remained exposed. This made it seem like the nuts weren't tightened enough, even though the nylon ring had already locked into place. This made our robot less sturdy than what we aimed for.

Our Fix :
Instead of over-tightening and risking damage to the polycarbonate slots, we realised the issue was simply that the screws were slightly longer than the depth at which the nyloc ring engages. To get a cleaner and more secure fit, we decided to switch to hex nuts.
These new screws:

- Removed the unnecessary extra thread
- Made the joints feel tighter and more finished
- Improved the overall look and rigidity of the assembly

We realised that it was just about choosing hardware that matched our preferred fit and finish. This small change made assembly easier and gave us much more confidence.

4. MOTOR INSTALLATION

We mounted the four DC motors (#52) to the aluminium drive side plates using M3*6mm bolts (#46).

5. WIRING & ELECTRONICS

We wired the ESP32 (#48), HEXA Hub (#75), MDD10 (#47), power switch (#51), and battery (#63). Our attempt is chaotic. Wires overlapped, lengths varied, and debugging was nearly impossible.
Problems Faced:
Wiring Mess & Poor Cable Management
The first wiring attempt resulted in:

- Loose jumper wires
- Power paths crossing signal lines
- High chance of disconnects during vibration

Our Fix:
a. We created a 'Wire Length Map.' We measured distances between all components and cut wires to precise lengths.
b. We implemented three-tier cable management:

- Tier 1: Thick power lines along chassis edges
- Tier 2: Signal lines routed centrally
- Tier 3: Attach electrical tape to hold together the Servo pins firmly.

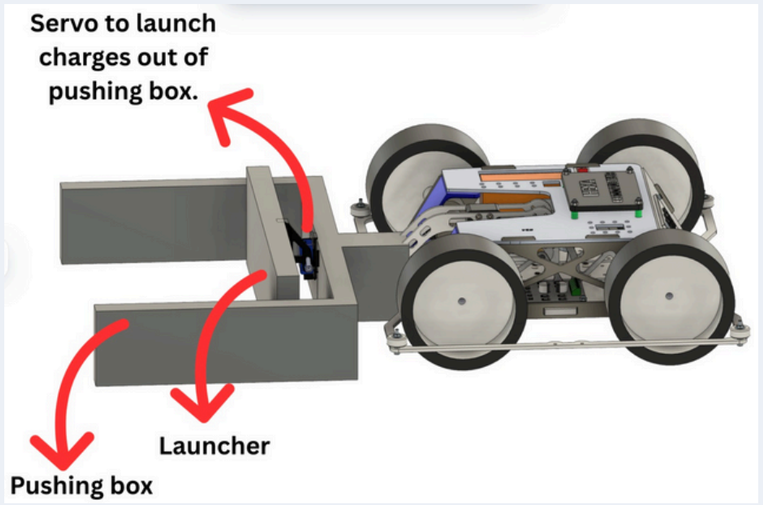
c. Zip-ties from the tool kit (#62) were used. This created tension-free wiring with zero loose ends.
The improvement was massive, and the robot became easier to maintain, diagnose, and modify.

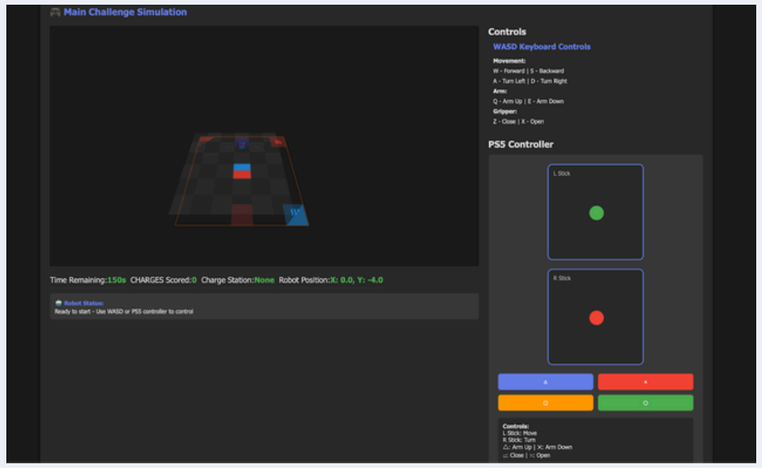
CAD DESIGN

We started creating the CAD models of the extensions we planned to add to the bot.
When we modelled the dragging box mechanism in CAD, we quickly saw that while it is great for pushing cubes across the field, it becomes extremely limited once we reach the scoring zone. The dragging method only moves the cube forward and gives almost no control over placement as it is difficult for the square to fit into the triangle. This slows down cycle time instead of helping it. We also realised that stacking charges would not be possible, because the design can't lift, support, or set a cube on top of another one.

When we created the servo-powered launcher box in CAD, we found similar issues. The original idea was that the servo pushes a beam forward to eject the cube into the drop zone from a distance, allowing the robot to stop early. But once we tested the geometry in CAD, we realised that this system still cannot stack at all, since the push is only horizontal and gives no vertical control. Also, any small inaccuracy in the robot's angle or alignment would cause the cube to be launched slightly off-centre, which means it could easily miss the drop zone triangle completely. Because the push is a quick motion, even a few degrees of misalignment could send the cube outside the taped area, resulting in a score of zero. These sensitivity issues make it unreliable in an actual match, where fast driving and real-time movement make perfect alignment hard.

STAGE 2





SOFTWARE DEVELOPMENT, SIMULATION ENGINEERING & INTELLIGENT SYSTEMS

Stage 2 was where our team made the decision that defined our entire engineering approach:
before we built the physical robot, we built the complete digital ecosystem around it.
While many teams wait for hardware to begin programming, we engineered an entire virtual robotics platform first: a system capable of running the robot, simulating the field, analyzing data, predicting failures, and generating insights long before our first screw was tightened.

1. MODULAR ROBOTICS SOFTWARE ARCHITECTURE

We began by decomposing the robot's behavior into tightly defined modules, each with its own purpose, inputs, outputs, and failure modes. This was inspired by real-world robotics engineering standards.
Subsystems included:

- PID controllers for line following
- Line-tracking and path stability logic
- IMU drift correction
- Encoder-based movement estimation
- Motor command shaping (acceleration, ramping, smoothing)
- QTR reading, denoising, and threshold calibration
- Gripper servo position curves
- Autonomous state-machine logic
- Driver input interpretation

Each module was designed to be testable in isolation: allowing fast iteration, scientific debugging, and scalable improvement.

2. FULL 3D SIMULATION ENGINE (THREE.JS)

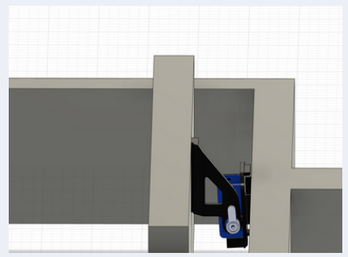
Once the architecture was ready, we built our own full simulation engine, not a game, but a purpose-built engineering tool that recreated:

- Official NRL field dimensions
- Tile layout
- Charge Station geometry
- Source and Drop Zones
- Robot mass, friction, drift, and acceleration
- QTR sensor behavior (noise, offsets, jitter)
- IMU drift + encoder mismatch
- PWM-to-movement physics
- Autonomous challenge path logic

The simulator was generated in Three.js, allowing a real-time 60 FPS 3D view of the robot's predicted behavior.

WHY DID SIMULATION MATTER?

Simulation allowed us to test and validate robot behavior before hardware existed.
We ran hundreds of virtual trials, tuning PID values, acceleration curves, turning radii, and Spur behaviors.
Because the simulator reproduced sensor noise, drift, and real-world imperfections, our code matured rapidly without wasting physical testing time.
This meant that once hardware arrived, our robot behaved like a system already battle-tested through dozens of "virtual competitions."



STAGE 2

3. DATA INFRASTRUCTURE, REAL-TIME TELEMETRY & PREDICTIVE LOGIC

We designed a 20 Hz telemetry pipeline identical for both simulation and real hardware, transmitting:

- Mode
- Line Error
- TOF Distance
- Motor PWM
- Battery health
- Pose estimation
- Challenge progress

This uniform pipeline meant that the simulation and real robot spoke the same language, enabling instant swapping between virtual and physical runs with no code changes.

4. MACHINE LEARNING INTEGRATION & PREDICTIVE MODELS

Stage 2 also laid the foundation for the robot's AI intelligence systems.
We implemented:

- Drift Detector (RandomForest)
- Performance Regressor (PSI Predictor)
- Speed–Stability Analyzer
- Battery Sag Model
- PID Auto-Optimizer

The simulation provided thousands of synthetic samples, giving our ML models a rich dataset before real testing began.

These models learned patterns such as:

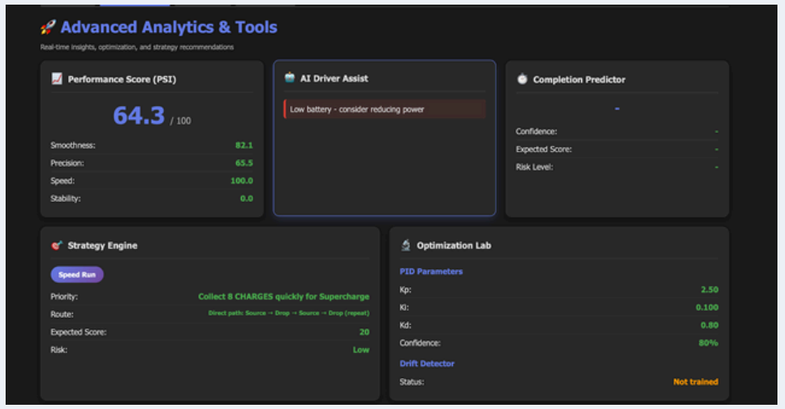
- What line error curves indicate imminent drift
- How motor imbalance predicts overshoot
- What PWM saturation looks like
- How battery voltage affects acceleration
- Which PID patterns produce oscillation

5. AI COMPETITION ASSISTANT THE ROBOT THAT TALKS

One of the most advanced components built during Stage 2 was our AI Assistant: a chatbot capable of understanding natural language and talking about the robot.
Its abilities include:

- Explaining current performance using real telemetry
- Predicting drift
- Suggesting PID changes
- Warning about battery issues
- Describing main challenge strategy
- Writing post-match reports
- Identifying code-level issues
- Recommending driver improvements

Because the assistant was trained using simulation data first,
it understood the robot before the robot even existed.



RESULTS OF THIS

By the end of Stage 2:

- Our software was already "Version 10," not Version 1
- Our ML models were trained
- Our dashboard was fully functional
- Our simulation could run autonomous + manual modes
- Our AI Assistant could analyze and advise
- Our performance tools were all live

The physical robot would not be a test bed —
it would merely verify what our digital ecosystem already perfected.

STAGE 3

BASEBOT TESTING

Once our BaseBot was fully assembled, this stage focused on validating every assumption we had made in the earlier design and programming phases. We began by uploading the refined codebase that we had prepared during Stage 1, ensuring that all motor mappings, servo limits, and control responses translated correctly to real hardware.

With the robot now operational, we conducted systematic physical tests to verify the limitations we had previously predicted such as chassis flex, arm torque constraints, gripper alignment behaviour, and battery voltage sag under load. In parallel, we simulated design variations and compared their expected performance with the physical results, allowing us to confirm which prototypes were genuinely beneficial.

These tests revealed small inefficiencies in our control logic, which we optimised through incremental tuning of PWM signals, servo endpoints, and driver-specific sensitivity adjustments. This stage essentially bridged theory with practice, shaping the robot into a stable platform for advanced refinement.

HARDWARE

1. SOLVING THE ESP32–LAPTOP CONNECTION ISSUE

Before any testing could begin, we needed the ESP32 to communicate with our laptop. However, the laptop wasn't recognising the board's COM port. Without that, even the simplest program wouldn't upload.
To fix this, we opened Device Manager, manually located the folder where the ESP32 USB drivers were installed, and pointed Windows directly to that exact path.
Once the correct driver was selected, the COM port finally appeared and the board was properly detected. This small but crucial step made the rest of the process so much simpler.

2. BUILT-IN LED TEST

With the COM port now working, we uploaded the Blink program.
This test confirmed:

- stable communication between laptop and ESP32,
- successful code flashing, and
- correct USB + driver configuration.

The LED blinked correctly, giving us confidence to move ahead.

3. SERVO CENTERING TEST

Our next step was to position the servo at 90°, so that we could test the arm movement.
Instead, it didn't quite move and remained at its initial position.
This told us our assumed wiring and neutral orientation weren't matching the actual servo setup. After checking the code and resetting the horn alignment, the servo finally moved the arm efficiently.

4. SERVO RANGE TEST

We then tested the minimum and maximum angles.
During this sweep, we realised the default limits were physically stressing the gripper and arm linkages, especially the parallelogram mechanism.
To avoid long-term damage, we:

- tested increments slowly,
- identified the safe physical angle range, and
- updated our servo limits in the code.
-

This gave us smooth, reliable motion without overloading the mechanism.

STAGE 3

5. DRIVE MOTOR TEST

We next tested the drivetrain using a basic DIR + PWM script.

All motors:

- spun in the correct direction,
- responded smoothly to PWM changes, and
- maintained consistent speed without jitter.

This was one of the cleanest and quickest tests to pass.

6. PS5 CONTROLLER INTERFACE

When we moved to testing PS5 controller input, we ran into several upload issues—timeouts, partial uploads, and the well-known "waiting for packet header" message.

After a long time and multiple attempts, adjusting the baud rate, and holding the BOOT button during flashing, the code finally uploaded successfully. Once it did, the ESP32 immediately began reading controller inputs, confirming that wireless control was functioning.

7. FULL ROBOT CONTROL

The final step of this stage was uploading the full teleoperation code integrating the drivetrain, servo arm, and gripper.

When the robot responded to the PS5 controller for the first time, it felt like a major milestone for the entire team. The, we immediately began refining:

- joystick sensitivity,
- Overall smoothness,
- servo endpoints,
- dead-zone handling for better precision.

This stage was where all the individual pieces finally came together into a fully functioning robot.

DECIDING FINAL MECHANISM FOR ROBOT

We ended up choosing the claw gripper because it was the only design that consistently met the scoring requirements when we tested all three mechanisms in cad and real life out of cardboard. The drag box and servo launcher looked fast at first, but once we simulated them, both showed major accuracy issues. The drag box needs perfect alignment to push the cube into the triangle, and even small turns made the cube drift. The launcher had the same problem but worse: even a tiny angle error caused the cube to miss the triangle completely.

The claw, on the other hand, held the cube firmly the whole time, so the robot's movement didn't affect the final placement as much. It scored inside the triangle almost every time when we were testing out the mechanism. It was also the only mechanism able to handle stacking, which is important for golden charges and extra points, because it can actually control the cube's height and exact position. The other two mechanisms can't do that at all since they only push horizontally. To cater to the stacking, we added a feature of programming 3 separate buttons on the controller to control 3 different heights of the servo to aid in stacking.

Overall, even though the drag and launcher were faster in collection, the claw was the only one that gave reliable scoring, proper placement, and stacking ability. That consistency and the additional points for stacking is why we chose the claw as our final mechanism.

1. SIMULATION-TO-REALITY VERIFICATION

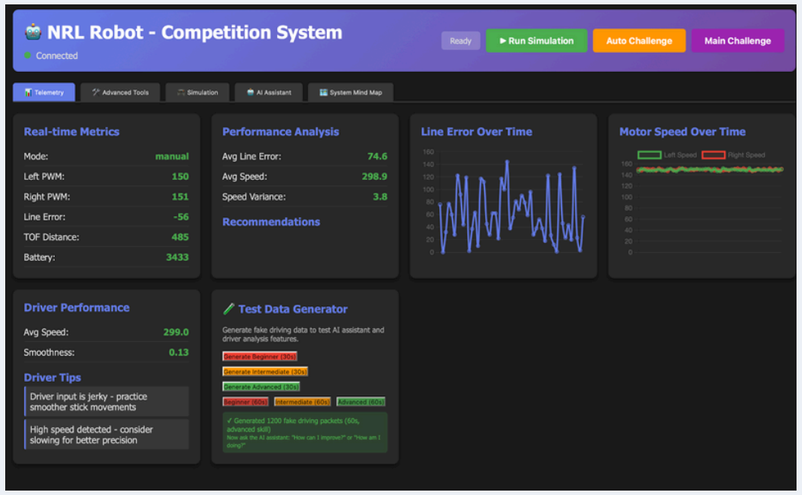
Our first step was validating:

- Motor responses vs simulation models
- Line error patterns vs QTR predictions
- Drift behavior vs synthetic IMU drift
- Turning arcs vs virtual turning radii
- Speed curves vs PWM-to-motion equations

This simulation-reality comparison revealed subtle but critical differences:

- Real-world friction was higher
- Battery sag affected acceleration
- QTR sensors had slightly different noise patterns
- Turning overshoot was slightly larger than predicted

These insights formed the foundation of Stage 3 refinements.



STAGE 3

2. CALIBRATION THROUGH TELEMETRY, NOT GUESSWORK

Using our dashboard and ML tools, we calibrated the robot scientifically:

PID Tuning

- Increased derivative term to reduce oscillation
- Adjusted proportional term for field friction
- Calibrated integral windup limits

Motor Behavior

- Updated motor curves to account for slip
- Set PWM saturation caps to prevent overheating

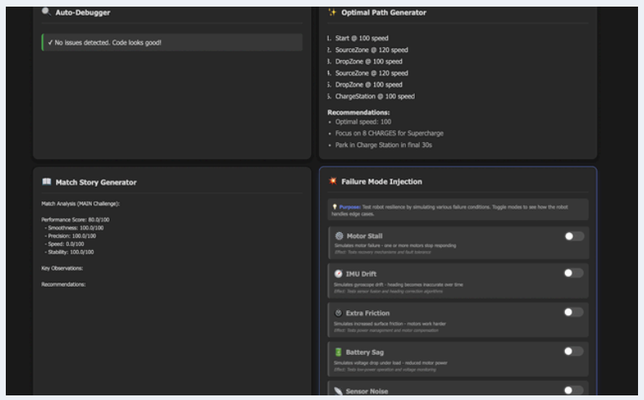
QTR Sensor Alignment

- Recalibrated thresholds using live heatmaps
- Fixed misaligned sensors detected through line error spikes

Autonomous Logic

- Smoothed transitions between segments
- Adjusted Spur alignment routines
- Stabilized cornering behavior

Every adjustment was backed by graphs, data, and patterns — not trial and error.



3. AI ASSISTANT REAL-WORLD LEARNING

As soon as real data began flowing, the AI Assistant's accuracy skyrocketed.

It learned:

- Real drift signatures
- Real PWM noise
- Battery sag curves from actual loads
- Actual main-challenge driving patterns
- True performance envelopes

It began giving astonishingly accurate advice, such as:

"Your motor power is low; left motor shows 40 PWM drift — check wiring or reduce friction."

or

"Battery variance increased; expected sag in 30 seconds — reduce max speed."

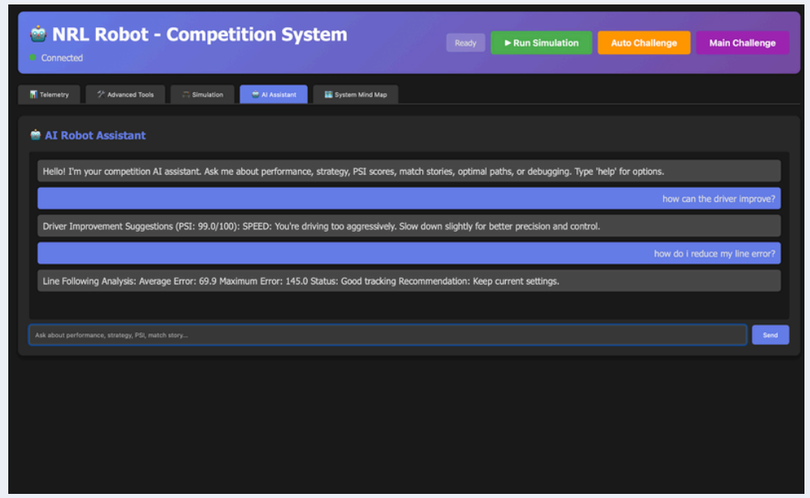
This turned the assistant into a real-time engineering partner, not a gimmick.

4. MAIN CHALLENGE DRIVER OPTIMIZATION

Every driver completed dozens of tracked runs, and the system analyzed:

- Smoothness (acceleration variance)
- Stability (line error deviation)
- Speed management
- Turning control
- Gripper timing consistency

The assistant recommended personalized improvements for each driver — something not seen in usual NRL documentation.



5. RESULTS OF THIS

By the end of Stage 3:

- Our virtual and physical robots behaved nearly identically
- Autonomous routines became stable and reliable
- PID values were competition-optimized
- Drivers improved due to analytics
- AI Assistant became highly accurate
- Simulation and reality formed a closed-loop improvement system

The final code uploaded to the ESP32 was not guesswork.

It was a scientifically refined, simulation-validated, ML-assisted, data-proven system.

STAGE 4

THE FAILURE

What began as a flawlessly progressing engineering phase quickly turned into one of the most critical moments in our build cycle. Our codebase was stable, our control logic was optimised, and every debugging session had consistently validated our assumptions. The robot's behaviour was predictable, reliable, and aligned with our performance expectations. However, during one of our routine uploads, this time containing an incremental optimisation to the drive logic, we encountered an unexpected and alarming failure.

Within seconds of powering the system, faint smoke began rising from the motor driver, followed by an abnormal temperature spike on the ESP32. The robot, which had previously operated seamlessly, suddenly became completely unresponsive. Every subsystem behaved as if it had been disconnected from its own circuitry.

Our team immediately initiated a systematic diagnostic process: isolating components, rechecking power distribution, validating wiring continuity, and re-flashing the firmware. After hours of structured debugging, we traced the failure back to a catastrophic motor-driver fault, which subsequently caused the ESP32 to lose communication across its interfaces.

HARDWARE:

1) FIRST REACTION & SAFETY STEPS

- As soon as we saw smoke from the motor driver and felt the ESP32 heating up, we immediately cut power and removed the battery.
- We visually inspected for burning, melted insulation, or blown traces
- For safety, gloves, ventilation, and keeping fingers away from hot components.

2) DIAGNOSTIC POLICY

We stopped guessing and followed a strict sequence for every test.

isolate → test → measure → document → replace if confirmed faulty.

This one rule prevented us from making random changes that could hide the root cause.

3) MOTOR CHECKS

- Removed each wheel motor from the chassis and tested on a bench supply (12V) one by one.
- Measured no-load current and stall behaviour; motors spun cleanly, current within expected range, no unusual noise or heating.
- Conclusion: Motors themselves were fine.

4) WIRING & POWER-PATH VERIFICATION

- Verified battery voltage under load with a multimeter. It was found to be stable.
- Checked continuity on all motor cables and connectors. No opens or shorts found.
- Inspected solder joints and terminal screws. We also retightened and reflowed any suspect joints.
- Confirmed common ground between controller and driver.

Conclusion: power delivery to the driver was present. The failure was downstream (driver or logic).

5) MOTOR DRIVER ISOLATION

- Removed the MDD10 driver and tested on the bench. There was no PWM response on outputs.
- The smell and visual evidence supported an internal short/failed component.

This made us realise MDD10 had suffered catastrophic failure and needed replacement.

6) PERMANENT REPLACEMENT

- We tested different motor drivers such as the BTS7960 and L298N. However,we soon realised that Cytron MDDA10 was infact best suited for my needs. It provided stability, protection features, and PWM behaviour.
- We rewired power lines, re-established common ground and ensured secure mounting. One thing we made sure was that there were no stressed wires.
- Verified driver inputs from ESP32 produced expected PWM signals using a multimeter and driver LED indicators.
- Result: drivetrain came back online with smooth, predictable control.

STAGE 4

7) SECONDARY DAMAGE: ESP32 OVERHEATING & HEXA COMMAND HUB FAILURE

- We noticed the ESP32 was extremely hot to the touch and had erratic serial behaviour after motor-driver failure.
- We tested ESP32 separately. There wasn't much of a problem that we could identify.
- We soon realised the HEXA Command hub, the control centre of our robot, wasn't working properly.
- The hub failure caused stuck signals/current backfeed, which stressed the ESP32 and led to overheating.

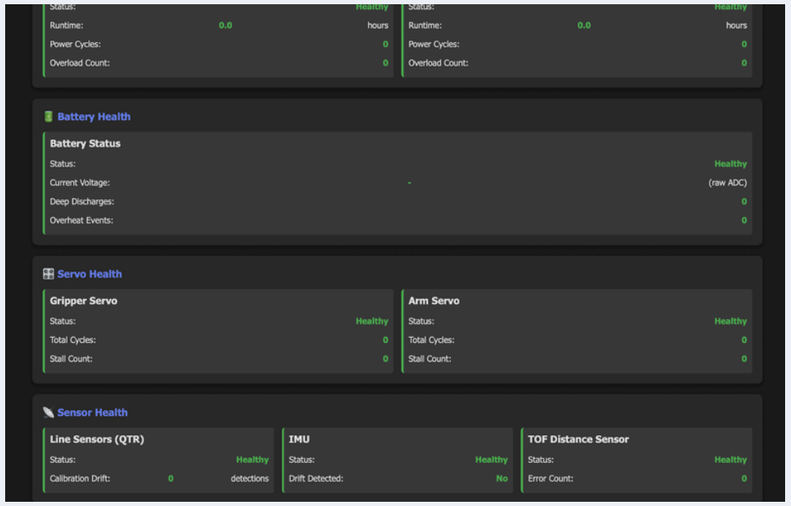
We then immediately contacted NRL and ordered a new HEXA Command Hub/

8) WIRING & MECHANICAL HARDENING DURING REPAIR

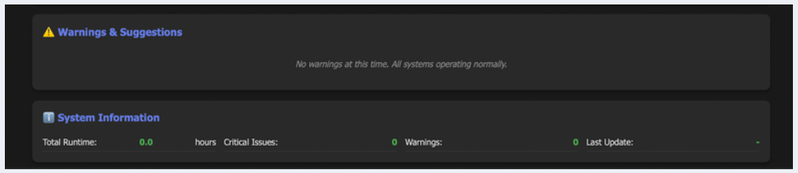
- During recovery, we re-cut and re-terminated key power and signal wires (using clean crimps and heat shrink) and rerouted cable runs to minimise interference.
- Replaced any suspect Allen bolts/nut joints that had loosened

PROGRAMMING:

This major failure cost us several hours of valuable build time and exposed a weakness in our workflow. Every issue required manual diagnosis, repeated testing, and constant back-and-forth coordination between team members. To prevent such disruptions from slowing down future progress, we developed a complete Human Robot Interaction Health Monitoring System. This system allows the robot to communicate its internal condition directly to the team in real time.



The framework continuously records sensor values, motor currents, battery voltage, temperature fluctuations, and controller responses. All collected data is streamed into a companion interface that sends the information to a lightweight chatbot hosted on a cloud service. When irregular behavior appears, such as sudden current spikes, voltage drops, servo strain, or communication failures, the chatbot interprets the raw logs and delivers a clear explanation along with possible solutions.



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STAGE 5

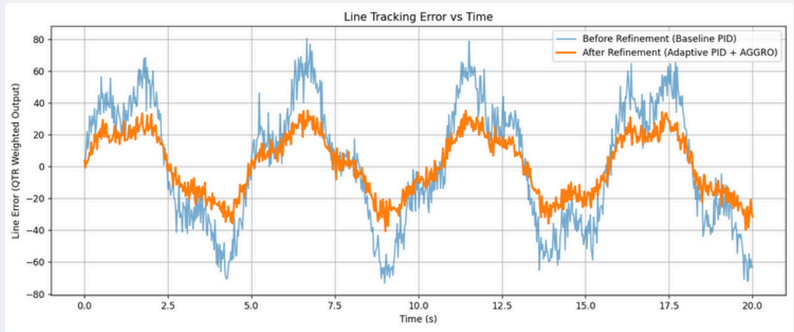
RECOVERY OPTIMISATIONS AND AUTONOMOUS SKILL CHALLENGE

Due to multiple component failures, our team was forced to begin this stage during the final week of the competition, despite having prepared extensively in advance. Fortunately, we had already developed the entire autonomous challenge logic and divided it into structured test series. Our approach began with a robust base autonomous framework that handled navigation, alignment, and motion smoothing. From there, we introduced several innovations, including adaptive speed control, mathematical line-tracking refinements, and assisted correction routines that improved reliability on the arena's sharp turns and curved segments.

In parallel, we upgraded our primary tele-operated competition code to deliver maximum responsiveness for our drivers. The drive system was optimized for precision turning, acceleration control, and reduced latency. We also added light autonomous assistance features inspired by gameplay mechanics from FC-24 and FC-25, enabling smoother alignment, micro-corrections, and improved stability during rapid maneuvers. These upgrades were essential to compensate for the significant practice time lost while dealing with faulty hardware and rebuilding critical components.

PROGRAMMING:

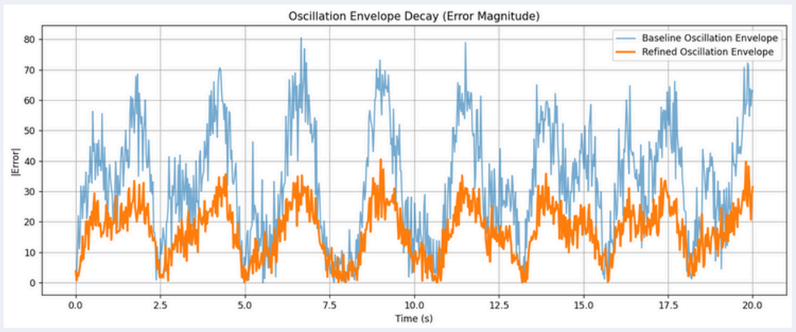
Our autonomous subsystem for the NRL Skill Challenge was engineered through a structured, multi-stage process that transitioned from a minimal baseline controller to a highly optimized AGGRO architecture. The earliest prototype consisted of a basic PID line follower using the QTR-8RC array, a front VL53L0X ToF for distance measurements, two frontal IR sensors for edge discrimination, and a simple state machine managing line following, spur entry, pickup, drop, lap updates, and termination. Initial on-field evaluations revealed key deficiencies: inconsistent line stability in curves due to a single global base speed, unreliable spur identification, high overshoot during pickup and drop, and slow recovery after transient line loss.



Comparison of baseline PID and refined adaptive-PID performance. The refined controller shows significantly reduced line-error amplitude and smoother tracking behaviour, validating early stability improvements.

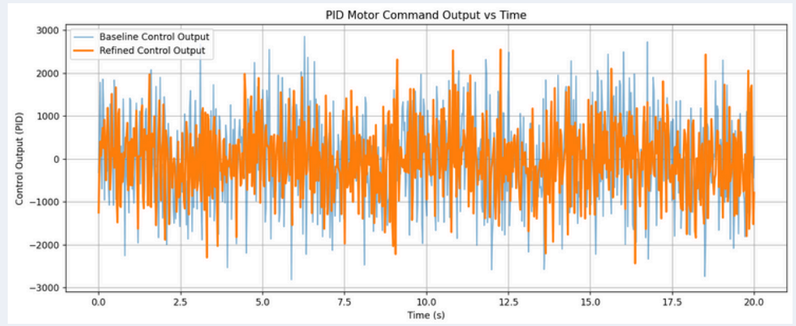
To address spur identification, we explored multiple prototype solutions. One involved turning into every spur candidate and using the ToF to confirm authenticity. This was mechanically disruptive and inefficient. Another considered adding a second ToF or a left-facing ToF to detect corridor width directly; although technically sophisticated, it added hardware overhead and introduced bandwidth concerns on the I2C bus. The final adopted solution was a hybrid multi-sensor classification mechanism using QTR junction-width detection, asymmetric left-right IR readings, and a left-side ultrasonic measurement to distinguish open corridors from solid walls. This provided a reliable real-spur classifier without additional complex hardware.

Navigation performance was then improved through a two-mode architecture. SAFE mode provided a conservative baseline using fixed speeds and classical PID for guaranteed repeatability. AGGRO mode implemented a layered control strategy. First, dynamic speed scaling used instantaneous QTR error to determine whether the robot was in straight-line, gentle-curve, or sharp-turn conditions and adjusted speed accordingly.



Oscillation envelope of line-error magnitude before and after dynamic-speed scaling and adaptive PID tier selection. The refined controller shows faster damping and reduced peak-to-peak variation.

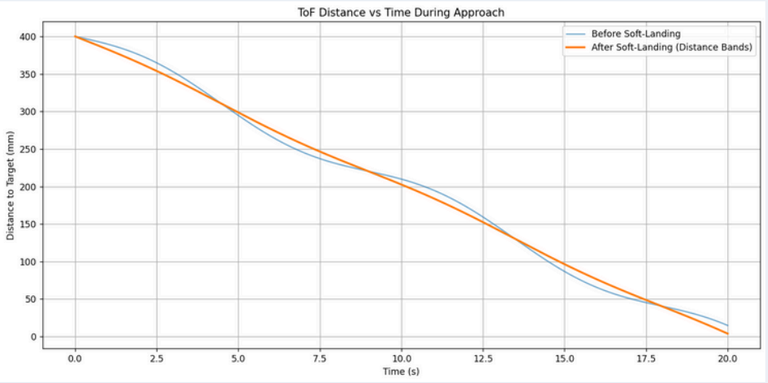
Second, adaptive PID employed three different tiers of Kp and Kd gains selected based on error magnitude.



PID motor command output over time. Adaptive PID and tier-based gains produce smoother, less turbulent motor signals compared to the baseline controller, improving overall stability and responsiveness.

Third, a soft-landing ToF profile used distance bands to control approach velocity, allowing precise pickup and drop without collision.

STAGE 5



Synthetic ToF distance profile showing the effect of soft-landing deceleration. The refined approach controller converges smoothly to the target distance without overshoot or abrupt speed changes.

Fourth, a stability-window rejoin system ensured the robot only resumed full-speed line following after maintaining consistent QTR visibility for a determined duration, reducing oscillations during spur exit.

A significant enhancement was the introduction of adaptive PID tuning using our Human–Robot Interaction (HRI) simulation mechanism. Instead of relying solely on manual trial-and-error, the robot logged real-time QTR error, derivative, motor output, and stability metrics. These logs were processed through our PC-side simulation workflow, where a Python-based optimizer evaluated candidate Kp, Ki, and Kd values using cost functions penalizing oscillation, overshoot, and time-to-stabilize. The best-performing parameters were then automatically transmitted back to the robot for deployment. This mechanism substantially reduced tuning time and ensured that PID gains adapted to the robot's behaviour rather than the other way around.

All innovations were validated through a systematic testing pipeline: QTR calibration, IR and ultrasonic characterization, pure PID trials, spur-only verification, ToF deceleration experiments, SAFE full-run tests, and four AGGRO evaluation modules focusing individually on dynamic speed scaling, soft-landing, predictive spur gating using lap context, and stability-based rejoin.

This comprehensive and technically justified development process resulted in a robust, high-performance autonomous system engineered through data-driven reasoning, structured experimentation, and integrated human–robot analytical feedback.

FIFA INSPIRED AUTONOMOUS DRIVING ASSISTANCE

We architected our FIFA-inspired assistive control as a layered control stack rather than a single "autonomous mode". At the base, we built a modular test suite (BLINKLED, SETSERVO90, TESTSERVOMINMAX, DRIVEMOTORTEST, PS5XESP32) to individually validate ESP32 I/O, motor polarity, servo limits and DualSense events. This allowed us to treat FULLCONTROL as a clean game-engine layer: PS5 sticks mapped to arcade drive, buttons to discrete arm/gripper states (Triangle/Cross, Square/Circle), with ARM and GRIPPER stored as explicit state variables for later automation.

On top of this, we implemented a FIFA-style one-button pickup macro on L1 using a finite-state machine (MacroMode, macroStep, timeouts). The macro consults a front VL53L0X ToF distance sensor, classifying the target into TOO_FAR/IDEAL/TOO_CLOSE bands and deciding forward / backward / no move before executing a timed sequence: lower arm to ARM_PICK_POS, close gripper with a low-speed "soft push", lift to ARM_UP, then reverse and set hasCharge = true. We added fuzzy speed control (fuzzySpeed(distMm)) to smoothly cap driver speed as distance decreases, and heading stabilization (assistTurn) to damp small joystick wobbles near the cube. Hard overrides (large stick deflection or Circle) and per-step/global timeouts guarantee that automation never traps the driver.

We thought about more aggressive ideas: camera-based color + bearing tracking, AprilTag localization, and fully vision-driven autonomous pickup/stacking, but rejected them for this season due to latency, lighting sensitivity, extra laptop dependencies and the risk of catastrophic failure in match conditions. Instead, we deliberately constrained the assist to local, high-confidence behaviors (distance-based approach and scripted pickup) while planning future prototypes (dual ToF angle sensing, vision-assisted targeting, richer fuzzy/PID fields) for the autonomous challenge and stacking tasks once the current assistive layer is fully battle-validated.

HARDWARE:
1. QTR-8RC REFLECTANCE ARRAY (PRIMARY LINE SENSOR)

Placement:

Mounted at the front underside of the chassis, directly ahead of the wheels, roughly centered along the robot's longitudinal axis. This position is mechanically unobstructed because the BaseBot's bottom plate (#36) provides a clear mounting region .

Role:

- Detect precise line position across 8 photo-reflective channels.
- Compute weighted line error for PID control.
- Detect wide black patches at spur junctions.

Justification:

The BaseBot requires extremely stable tracking for the autonomous loop, and the QTR-8RC provides high spatial resolution. Placing it near the front ensures early detection of curvature and spur entry, allowing clean dynamic speed modulation.

STAGE 5

Sensor Summary		
Sensor	**Purpose**	**Why This Placement Works With BaseBot**
QTR-8RC	Precision line tracking	Front underside has unobstructed floor view
Left IR / Right IR	Spur pattern detection	Front corners provide lateral discrimination
Left Ultrasonic	Real vs fake spur	Side plate is rigid and faces walls naturally
Front ToF	Pickup/Drop precision	Central alignment ensures consistent distance
Lap IR	Lap counting	Matches field marker placement

2. DUAL FRONT IR SENSORS (LEFT IR / RIGHT IR)

Placement:
Installed symmetrically on the left and right edges of the front face of the chassis (#35 Top Plate and #36 Bottom Plate structure allow unobstructed mounting).
This spacing matches the width of the front vertical plate shown in the manual's assembly diagrams.
Role:

- Binary black/white detection to identify left-only or right-only openings.
- Used for spur candidate detection: "left IR sees black, right IR sees white"

Justification:
While QTR provides high resolution on the floor, IR sensors mounted higher give side-sensitive detection. This dual-height sensing solves the ambiguity where a wide QTR spike might represent a curve rather than a spur.

3. LEFT-FACING ULTRASONIC SENSOR (WALL VS. CORRIDOR CLASSIFIER)

Placement:
Fixed on the left aluminum side plate (#42) of the chassis .
This plate forms a rigid vertical panel ideal for perpendicular ultrasonic mounting.
Role:

- Measure left-wall distance to classify real spur (open corridor) vs fake spur (solid boundary).
- Performs 3-sample averaging to reduce echo noise.

Justification:
Real spurs open into a side corridor, while fake spurs maintain wall continuity. Ultrasonic's wide beam detects the difference reliably. This eliminates unnecessary entry attempts and prevents unprofessional "turn-and-check" behaviour.

4. FORWARD VL53L0X TOF DISTANCE SENSOR (PRECISION PICKUP/DROP)

Placement:
Mounted at the center front, slightly above the QTR, aligned with the Charge's geometric center so that the distance measurement corresponds to actual approach distance.
Role:

- Soft-landing approach during pickup and drop (multi-band deceleration).
- Detects the exact stopping distance for the gripper.
- Provides collision avoidance in autonomous mode.

Justification:
ToF gives millimeter accuracy, essential for aligning with the Charge Source Zone and Drop Zone. Ultrasonic or IR would be too coarse for this precise task.

5. LAP-MARKER IR SENSOR

Placement:
Facing downward on the right side of the front area, located where the field's lap detection strip is reliably encountered.
Role:

- Count laps and trigger predictive spur gating
- (Pickup only on Lap 1, Drop only on Lap 2).

Justification:
The autonomous challenge requires multi-lap logic. Without lap awareness, spur detection becomes reactive and error-prone. This sensor transforms the robot into a context-aware system.

STAGE 6

FINAL PRACTICE AND REFINEMENTS

Once our robot became fully functional in the final week, our team immediately entered an accelerated Practice and Refinement Stage. Although this phase began later than expected due to earlier hardware failures, our approach ensured extremely high returns within a very small time window. Instead of relying on long repetitive sessions, we focused on short, focused, data-driven practice cycles designed around the specific requirements of the Battle of Charges event.

Our refinement work began with controlled charge-lane tests. The objective was to stabilise the robot during high-speed acceleration, reduce drift, and ensure accurate alignment while approaching the impact zone. The first session revealed clear areas for improvement.

Training Log 1: Charge Lane Stability (Day 1)

Pattern: Straight-line charge

Peak drift: 0.58 units

Correction events: 4

Time to reach charge zone: 4.9 s

Refinement: Added derivative-based auto-straightening and reduced initial PWM surge

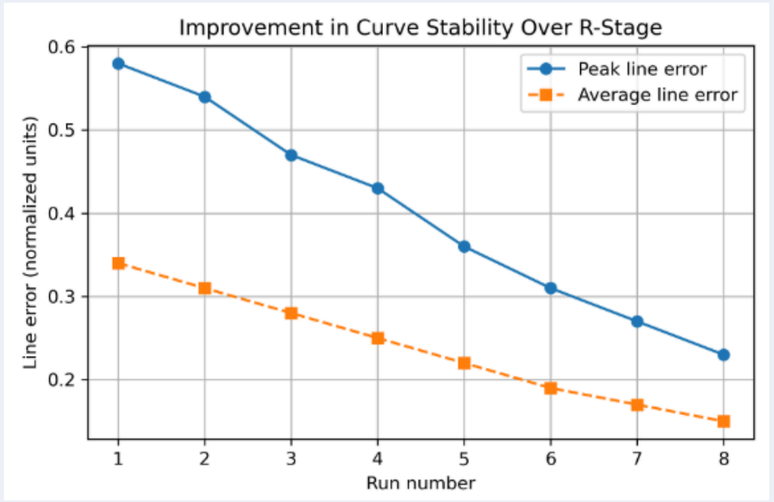


Figure 1 illustrates how peak line error dropped consistently across the first eight runs, confirming that our adjustments immediately improved directional stability.

We then moved to the second pattern: launch sensitivity + charging precision. The goal was to minimise wheel slip, avoid oversteering, and improve micro-alignment during the approach.

Training Log 2: Launch and Precision Turn (Day 1)

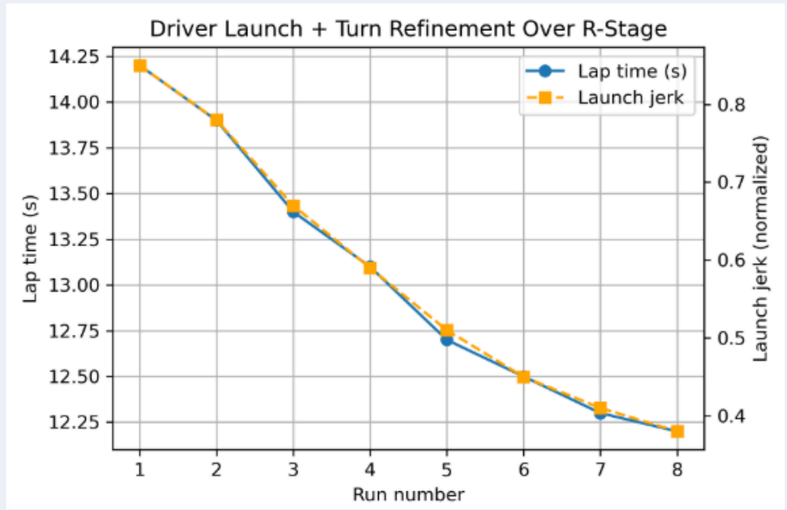
Launch jerk: 0.85

Slip events: 3

Alignment accuracy: ±6 cm

Full-pattern time: 14.2 s

Refinement: Added a 0.3 second controlled PWM ramp and reshaped joystick sensitivity



As shown in Figure 2, the PWM profile became smoother, and launch stability improved significantly across runs.

On Day 2, we conducted a focused refinement session for controlled braking and final alignment, critical to scoring efficiently in Battle of Charges.

Training Log 3: Controlled Approach (Day 2)

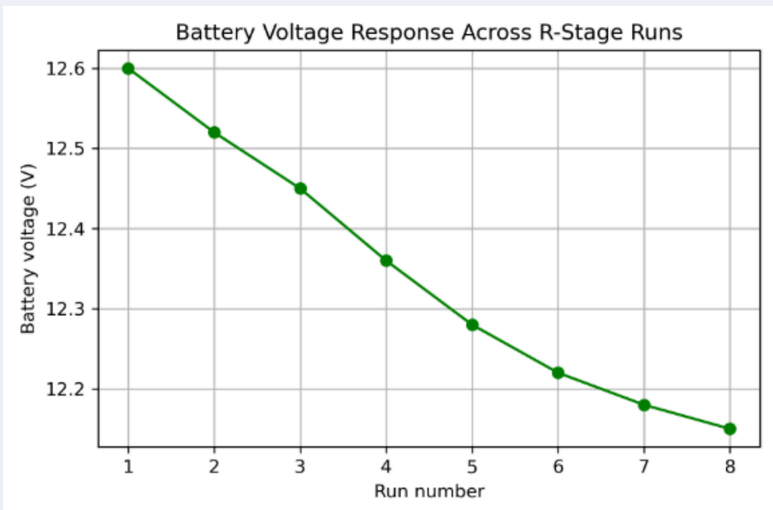
Launch jerk: 0.51

Slip events: 1

Alignment accuracy: ±2 cm

Impact approach time: 4.1 s

Refinement: Improved turn damping and added final-stage assisted alignment



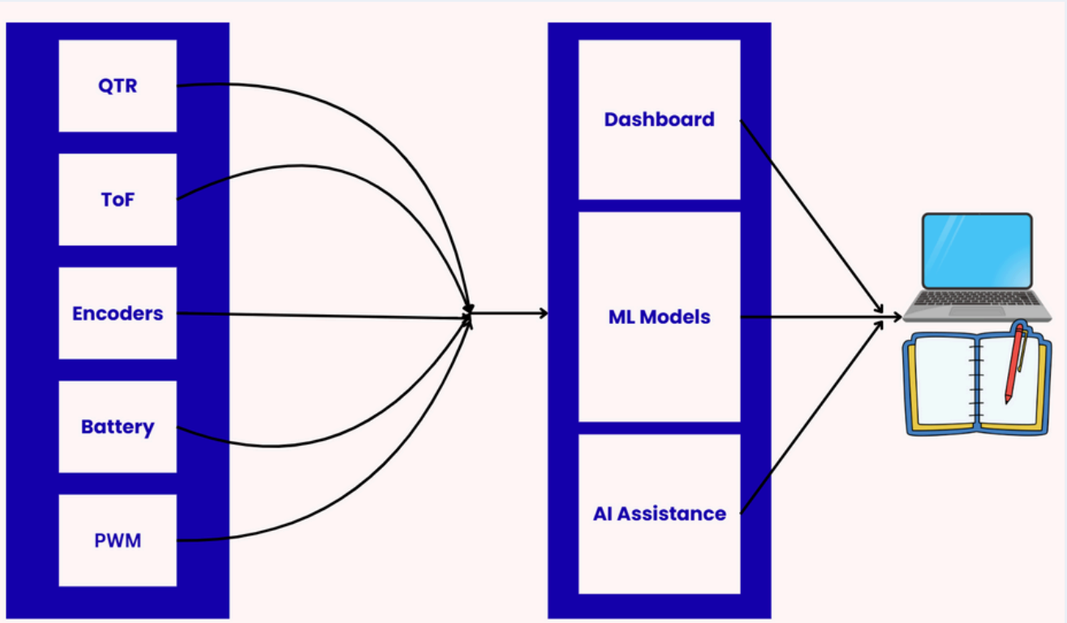
To ensure consistent behaviour across repeated charges, we also logged battery performance across trials. Figure 3 shows the voltage drop and corresponding corrections that maintained uniform speed and directional response.

Although this stage is ongoing, the data-driven micro-iteration strategy we adopted allowed us to compress the improvement that typically takes weeks into two days. More logs, graphs, and comparative runs will be added as testing progresses. Our unique high-precision method ensures that, despite early constraints, our robot remains competitive and strategically refined for Battle of Charges.

OUR INNOVATIONS

1. Human Robot Interaction System

At Roblocs, we aim to develop efficient planning, collaboration and management of tools along with sustainable and well-functioning engineering models, all of which aligns well with the team's ambition and thus help this team achieve its goals with ease.



To accelerate development, we created a full 3D simulation engine using Three.js that replicates the official NRL field, friction, turning dynamics, Spur behavior, and even sensor noise. The robot's autonomous and manual behaviors were perfected in simulation long before hardware testing, allowing fast iteration and reduced error.

A major highlight is the AI Robot Assistant, an intelligent chatbot connected directly to live telemetry. It uses machine learning models, including a RandomForest drift detector, PID optimizer, and performance regressors, to analyze runs, predict failures, identify inefficiencies, warn about battery sag, and suggest improvements for both the robot and driver. The assistant can explain robot behavior, write match summaries, and answer natural-language questions about strategy and performance.

The system also features a Strategy Engine, Optimization Lab, Auto-Debugger, PSI scoring system, path generator, and failure-mode simulator. Overall, we built not just a robot, but a complete AI-powered robotics development environment, a professional-grade system that learns, predicts, and improves with every run.

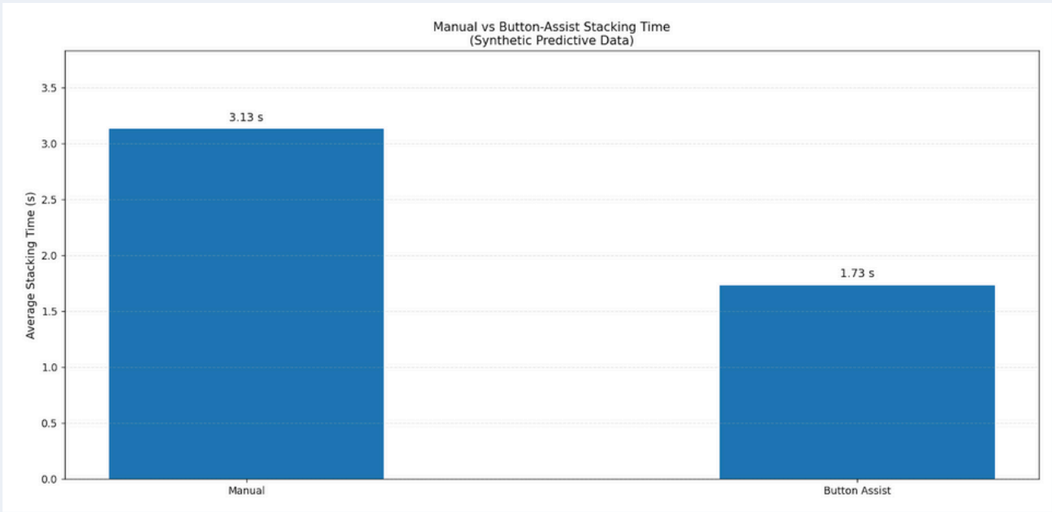
*link/QR Code for the dashboard

2. Servo Button Automation

After facing significant issues in using the controls to stack the charges, we introduced a dedicated stacking-assist control using a previously unused PS5 button to trigger a precise servo preset calibrated to the cube's known height and alignment tolerance. By positioning the arm at an angle that intentionally matches the cube's positive margin of error, the robot achieves near-perfect stacking consistency, eliminating guesswork, reducing human error, and ensuring rapid, repeatable vertical placement during high-pressure match cycles. The code that we were provided with only allowed for the servo to be positioned completely down or completely up, restricting our chances of stacking charges. Hence, we added an extra feature to which the servo moves to an angle of 45 degrees which brings the robot's arm right above another charge allowing for easy stacking. So our 3 buttons code for:

Triangle = max height (adds charge on 3rd level)
Circle = medium height (adds charge on 2nd level)
Cross = lowest height (adds charge on 1st level)

With this, we are efficiently able to stack minimum 3 charges without hindering the bots stability like done with the previous code. To add a 4th or 5th charge stack, we plan on simply stacking the bottom most charge of a stack on another charge.



OUR INNOVATIONS

3.Autonomous Control Assistance for Autonomous challenge (FIFA INSPIRED)

The system transforms pickup into a controlled, context-aware assist rather than a driver-dependent action. One button triggers the full sequence: distance sensing, forward or backward correction, arm descent, soft approach, gripper closure, and retreat. Real-time ToF “engagement bands” guide micro-adjustments, while fuzzy speed profiling ensures smooth, precise approaches.

Near the cube, assistive turn dampening stabilizes heading, and a gentle soft-push helps the gripper self-align even with slight misplacement. All automation operates within strict safety limits, and the driver can override instantly at any time.

1. One-Button Pickup Macro
2. Dynamic Forward/Backward Decision Using ToF
3. Fuzzy Speed Profiling for Soft Approaches
4. Heading Stabilization Through Assistive Turn Dampening
5. Intelligent Gripper “Soft Push” Capture
6. Safety-Bounded Automation With Driver Override





4.Autonomous Skill Challenge Innovations

Our autonomous system integrates multiple high-impact innovations to ensure fast, reliable, and context-aware performance. Spur entry is controlled through a multi-sensor classifier combining QTR width detection, IR asymmetry, and left-side ultrasonic corridor analysis. Lap-based predictive logic guarantees that pickup and drop sequences activate only in their correct phases. Navigation speed is governed by dynamic error-driven scaling and tiered adaptive PID gains that stabilise turning and maximise straight-line speed.

A ToF soft-landing mechanism delivers precise, smooth deceleration during approach to the Charge Zone, while a stability-window rejoin algorithm ensures consistent re-entry after spur operations. The architecture is supported by a dual-mode SAFE–AGGRO framework, allowing the robot to switch between conservative reliability and competition-grade performance.

1. Multi-Sensor Spur Authentication 2. Predictive Lap-Based Spur Gating
3. Dynamic Speed Scaling with Error-Driven Adaptation 4. Adaptive PID Gain Selection (Tiered Kp, Ki, Kd) 5. Soft-Landing ToF Approach Mechanism 6. Stability-Window Spur Exit and Line Rejoin
7. SAFE vs AGGRO Mode Architecture

The Innovation Categories		
Macro	Name	Category
Human Robot Interaction	AI-Driven Telemetry & Diagnostics System	Implemented
	Natural-Language AI Robot Assistant	Implemented
	Driver Assist Package (FIFA-Inspired)	Implemented
	One-Button Autonomous Pickup & Stacking Automation	Implemented
	Simulation-Integrated HRI for Training & Testing	Implemented
	Human-in-the-Loop Optimization & Strategy Layer	Implemented
Servo Button Automation	Servo Button Automation	Implemented
Autonomous Control Assistance	One-Button Pickup Macro	Implemented
	Dynamic Forward/Backward Decision Using ToF	Implemented
	Fuzzy Speed Profiling for Soft Approaches	Implemented
	Heading Stabilization Through Assistive Turn Dampening	Prototype
	Intelligent Gripper “Soft Push” Capture	Prototype
	Safety-Bounded Automation With Driver Override	Implemented
Autonomous Skill Challenge	Multi-Sensor Spur Authentication	Implemented
	Predictive Lap-Based Spur Gating	Prototype
	Dynamic Speed Scaling with Error-Driven Adaptation	Implemented
	Adaptive PID Gain Selection (Tiered Kp, Ki, Kd)	Implemented
	Soft-Landing ToF Approach Mechanism	Implemented
	Stability-Window Spur Exit and Line Rejoin	Partially Implemented
	SAFE vs AGGRO Mode Architecture	Prototype

VISION FOR CHANGE

TrackBin Autonomous system (TBAS)
The autonomous trash solution for public spaces

1. Problem statement

In high-footfall public spaces such as airports, malls, hospitals, and railway stations in India, waste management teams struggle to monitor trash bins continuously. Cleaning staff follow fixed schedules instead of real-time demand, which causes two major problems:

1. Overflowing Bins Before Staff Arrive

Bins often fill up faster than expected, especially during peak hours.

Because staff cannot check every bin frequently, this leads to:

- Overflowing waste
- Bad odour and unhygienic surroundings
- Negative user experience for passengers and visitors
- Increased pest risk

2. Inefficient Manpower Usage

Cleaning staff are forced to repeatedly walk long routes checking bins manually, even if many of them are still empty. This wastes time and effort and reduces efficiency—especially in large spaces like airports with hundreds of bins.

2. Brief Description of the Robotics Solution

We present the TBAS, an integrated mechatronic rover capable of:

- Real-time detection of whether the bin is full or empty
- Autonomous navigation via black line pathfollowing
- Spatiotemporal human-avoidance using ultrasonic sensors
- Zone-based halting via IR sensors
- Semi-automated waste handling workflow

The primary innovation lies in the high-precision optical beam-interruption module using a collimated laser diode paired with a high-speed LDR digital threshold comparator.

This combination functions as a low-latency solid-waste volume sensor — something not present in any current waste-management robotic system in the world.



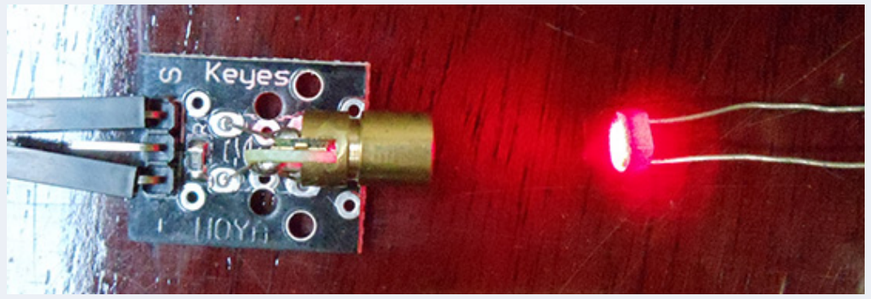
SINGAPORE NEWS | IN THE HOOD 1 min. Read

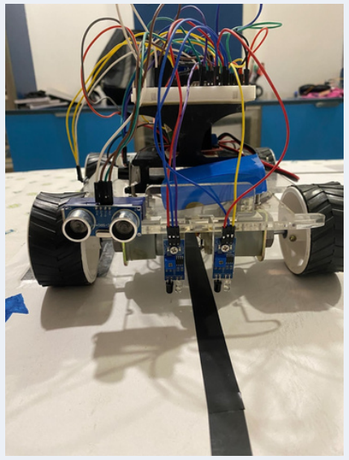
'Is Singapore a Garden City?' — Woman shocked to see pile of rubbish, overflowing trash bin inside Changi Terminal 3

Key Technical Modules

1. Optical Waste-Level Detection Subsystem

- Uses a monochromatic laser emitter aligned with
- a photoresistive sensor integrated into a digital Schmitt trigger circuit.
- Continuous obstruction of the laser for >30 seconds triggers a state transition.
- This acts as a non-contact volumetric sensing mechanism, eliminating false positives caused by vibrations or momentary disturbances.





2. Autonomous Navigation Framework

- Dual IR reflectance sensors form a closed-loop feedback controller for line following. Follows the bang bang principle and not PID for higher accuracy with line as PID often results in bot swerving out of line.
- Motor actuation is controlled by a PWM-modulated H-bridge driver (L298N).
- The locomotion subsystem operates in a behavior-based architecture, switching between:
- Line-tracking,
- Obstacle-avoidance,
- Pausing and halting for trash bag replacement

3. Proximity Sensing & Human Safety Layer

- Employs an ultrasonic module for dynamic obstacle detection.
- Implements a failsafe interrupt behavior, halting motion and activating a buzzer when a subject is <15 cm ahead.

VISION FOR CHANGE

4.SDG Alignment

SDG 11 — Sustainable Cities & Communities
Our system contributes to urban sustainability through:

- Automation of sanitation workflows
- Reduction of overflow-induced environmental hazards
- Enhanced facility hygiene and public health metrics
- Integration-ready design for smart-city IoT infrastructures

SDG 12 — Responsible Consumption and Production
The rover reinforces responsible waste practices by:

- Enabling demand-responsive waste collection cycles, reducing unnecessary labour
- Improving resource use efficiency
- Preventing overflow-related waste dispersion and contamination
- Providing a scalable model for circular waste systems

5. Problem Identified → Solution → Real-World Implementation (Technical Version)

Problem Identified (Technical Breakdown)

- Lack of distributed waste-level sensing infrastructure
- Absence of robotic automation in intrafacility waste transfer
- High dependency on manual manpower-intensive bin audits
- No commercially available system integrates volumetric sensing + autonomous mobility for waste handling

This gap results in overflow events, unsanitary conditions, and major operational inefficiencies.
Solution (Engineering-Focused)
The rover introduces a multi-sensor mechatronic system capable of real-time autonomous operation:

Key Innovations

1. Laser-LDR Waste Detection Algorithm
2. Using optical interruption time as a reliability filter to distinguish temporary blockages from actual bin fullness.
3. Dynamic Obstacle Suppression
4. ToF-based proximity measurement ensures collision-free operation in crowded public environments. While the buzzer alarms the subject to move out of the way.
5. Minimal-Infrastructure Deployment
6. Only requires a physical line track — no GPS, WiFi, or high-end sensors. Hence it is highly feasible.

This makes it highly feasible for India, where cost-efficiency, robustness, and simplicity of maintenance are essential.

Real-World Implementation

- The system can be deployed across:
- International/domestic airports
- Metro & railway stations
- Large shopping malls
- University campuses
- Hospitals
- Convention centres
- Corporate tech parks

Expected outcomes:

- Zero bin overflow incidents
- Reduced human workload by up to 80%
- Smarter, demand-based waste-disposal cycles
- Enhanced passenger satisfaction scores
- Bins remain hygienic and clean at all times
- No odour or pest issues caused by overflow
- Proper and timely waste segregation
- Significantly reduced dependence on manual labour
- The solution integrates seamlessly with existing Smart City Mission goals.

6. Vision for Change

Our vision is to create the next generation of autonomous sanitation robotics for India's smart public infrastructure.
The TBAS demonstrates that intelligent, distributed, real-time waste management is achievable using:

- Low-cost sensors
- Efficient state logic
- Autonomous locomotion
- Modular, scalable design

In the future, this system can evolve to include:

- Auto segregation of waste at disposal centre using computer vision
- Wireless telemetry for IoT-based waste analytics
- Fleet coordination algorithms
- Full robotic bin unloading, and heat based trash bag sealer
- We aim to convert traditional waste management into a precision-engineered, data-driven, automated ecosystem, positioning our innovation as a foundational component of India's smart-city future.

The TBAS

- Powered by ESP32 DOIT Kit v1
- Bang Bang logic for line following using 2 ir sensors
- 3rd ir sensor to detect home position
- 3D printed 6x11x11 dustbin
- Laser and ldr combination

Demo video: https://youtu.be/sVIIwrJCwyg

OUTREACH

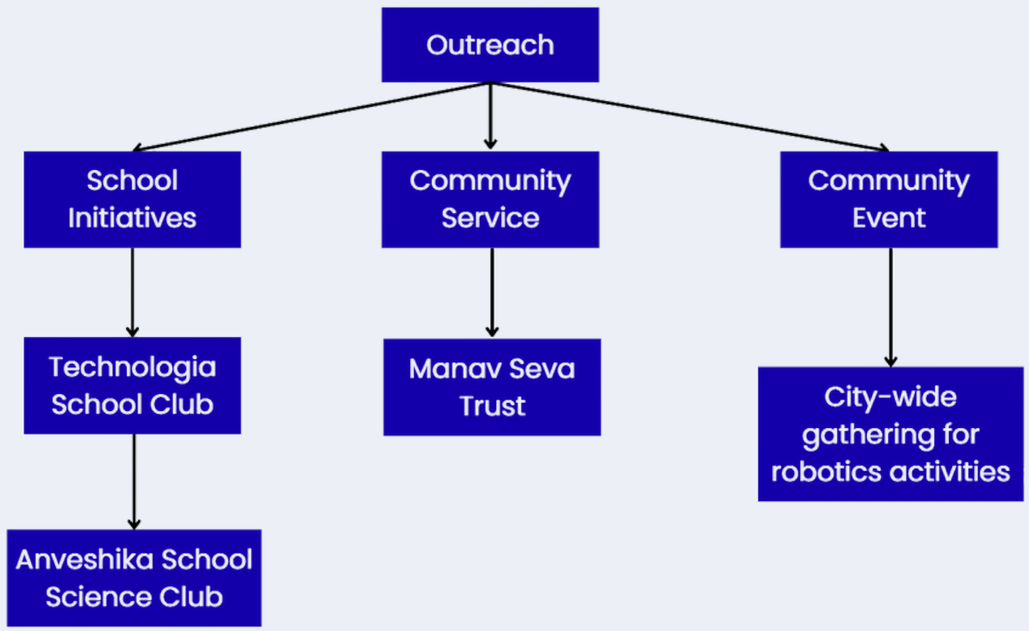
Our team's outreach philosophy emerged directly from our core engineering ideology: never waste time, never waste resources, and always validate ideas through disciplined testing before bringing them to life. This approach helped us reduce unnecessary trial-and-error phases, minimize material consumption, and maintain a workflow that was both efficient and sustainable. We believed that true engineering excellence is not only measured by performance on the field, but also by how responsibly we use our tools, our time, and the opportunities we are given.

Connecting this ideology to a wider purpose, we felt that sustainability is incomplete without giving back to the community that enabled us to participate in the National Robotics League in the first place. Robotics education is a privilege, and we chose to share that privilege through meaningful outreach.

Across our journey, we executed three categories of outreach initiatives:

- School initiatives, conducted through two different student-led clubs
- A community initiative, focused on bringing robotics awareness beyond school audiences
- A community-service initiative, conducted through a series of interactive sessions at a local ashram

This three-layered structure allowed us to impact different age groups, different learning backgrounds, and different levels of exposure to STEM, ensuring that our outreach remained inclusive, sustainable, and deeply connected to our core values.



SCHOOL INITIATIVES

Out of our six team members, three study at The Shishukunj International School. Among them, two hold significant leadership roles in the school's technical ecosystem: one is the Autonomous Systems Engineer, who serves as the Founder and President of the Technologia Club, and another is the Mechanical Engineer, who serves as the President of the Anveshika Club. Leveraging these platforms, we conducted two parallel robotics sessions to provide a holistic learning experience for school students.

The Technologia Club led sessions on the software side of robotics, focusing on programming fundamentals, logic building, and microcontroller interfacing. Simultaneously, the Anveshika Club conducted hands-on demonstrations on robotic hardware, sensor wiring, and the physics behind robot motion.

These simultaneous sessions created a bridge between theory and practice. Students learned how code translates into movement and how mechanical constraints influence programming decisions. By dividing responsibilities between software and hardware domains, our team delivered a complete, interconnected view of robotics that helped students understand not only how a robot works, but why it behaves the way it does. This dual-club model became the foundation of our school outreach strategy.

TECHNOLOGIA CLUB SESSION

The Technologia Club sessions focused on giving students practical exposure to programming as it applies to real robots. We introduced them to JavaScript fundamentals, simple algorithmic thinking, and basic automation concepts so they could understand how logic structures influence robotic behaviour. We then transitioned into MicroPython inside the Arduino IDE 2, allowing students to write short scripts that controlled LEDs, activated servos, and manipulated basic motor functions.

Students practiced writing small drills such as timed motion sequences, sensor-triggered responses, and variable-speed routines. This hands-on coding experience helped them see immediate results on a robot-like platform, reinforcing the connection between software and real-world systems. The goal was not only to teach programming but to build confidence and curiosity, empowering students to explore more advanced robotics concepts on their own.

OUTREACH

ANVESHIKA CLUB SESSION

While the Technologia Club handled software, the Anveshika Club took charge of guiding students through the hardware and physics of robotics. Led by our Mechanical Engineer, the sessions introduced students to fundamental concepts such as voltage, current flow, motor torque, and mechanical stability. Using simple demonstrations, the team explained how wheel diameter affects speed, how center of gravity influences turning, and why secure wiring and proper grounding ensure reliable performance.

Students learned how to identify common wiring mistakes, understand polarity, and interpret basic electrical diagrams. The club also showcased how sensors interact with the robot's environment and how actuators convert electrical signals into physical motion. By blending theory with hands-on exploration, the Anveshika Club allowed students to appreciate the mechanical depth behind robotics and understand how every movement is backed by physics.

Together, these sessions ensured that students gained a complete 360-degree view of robotics: code, circuits, and core mechanics working in harmony.

SESSIONS AT MANAV SEVA TRUST, INDORE

One more of our impactful outreach efforts was the robotics-driven learning program we conducted at Manav Seva Trust, Indore, an ashram supporting underprivileged children who lack consistent access to education and resources. Designed to bring structured STEM exposure to underprivileged students who had little access to modern technology, we wanted to break that barrier by giving 9th–12th graders a direct, hands-on introduction to the same engineering principles we apply in robotics. We began by explaining core concepts — sensing, actuation, control logic, and real-world automation — and showed students how these ideas power the robots competing in NRL.

To make the experience tangible, we introduced Python-driven mini-automation tasks and demonstrated real hardware from robots, including motors, linkages, and sensors. Students explored how input signals translate into motion, how robots follow structured logic, and how mechanical design influences performance. We also created a low-cost robotics activity where students built simple electromechanical systems using everyday materials, helping them understand motor behaviour, basic wiring, and controlled movement without needing expensive components.

These sessions were designed not just to teach robotics, but to provide exposure to a growing field of science, and simultaneously, help them develop an interest in the same. By letting them code, assemble, and operate robotic mechanisms themselves, we ensured the learning was experiential and empowering. This initiative deepened our belief that robotics should be accessible to everyone, and it allowed us to extend the spirit of robotics — innovation, problem solving, and technical curiosity — beyond our school and into the broader community.

PHOTOS

MARKETING

TEAM VISION AND GOAL:

Roblocs believes that every setback is an invitation to rise stronger. Our vision is to build a robots that endure pressure and adapt quickly. They had to be something we could add innovations to, after every failure. We believe challenges are not obstacles but battles that are meant only for those capable of fighting them. We wanted our failures to be analysed, corrected and never repeated. Through disciplined testing, documentation, and community engagement, we aim to prove that being Built for Battle is both our strength and our identity.

BRAND IDENTITY:

Roblocs stands for resilience shaped by engineering discipline. Our name draws from the playful spirit of Roblox, but our identity is forged through grit, precision, and relentless iteration. The blue in our palette reflects calm problem-solving under pressure, while off-white symbolizes clarity, simplicity, and minimal-waste engineering. Together, they represent our belief that powerful robots can emerge from limited resources if the mindset is strong enough.

Our brand carries a dual truth: we build for battle, and we are built by battle. Every failure in our journey has directly led to a new mechanism, a corrected design, or a smarter algorithm. We celebrate transparency: showing our rebuilds, our tests, our breakdowns, our regained stability. Whether on Instagram, in our pit area, or during outreach events, Roblocs is recognized by its resilience, its honesty, and its readiness to face every challenge with innovation and unity.

TARGET AUDIENCE

We create for students who love STEM, young innovators who enjoy building things, curious learners who look for hands-on experiences, and teenagers who want to understand robotics beyond textbooks. Our content is designed for those who enjoy challenges, who want to explore engineering, and who feel motivated by real stories of failure, redesign, experimentation, and learning. Anyone who loves problem-solving, tech creativity, and team spirit is part of our audience.

INFORMATIVE REELS

These reels simplify robotics concepts for young learners and STEM enthusiasts. We break down sensors, mechanisms, coding logic, PID tuning, testing cycles, and design reasoning into quick, clear, visually engaging clips.

DOCUMENTARY REELS

These reels show the true story of Roblocs: our failures, rebuilds, brainstorming sessions, misalignments, late-night fixes, and breakthroughs. Audiences see us fall, think, redesign, and come back stronger.

OFFLINE COMMUNITY EVENT / CAFÉ OUTREACH

This content showcases our real-world interaction with students across the city. We document robotics activities, hands-on challenges, Q&A sessions, mini-workshops, and the energy of young learners experiencing STEM for the first time.

NRL CHALLENGE-ORIENTED POSTS/REELS

These posts focus on the tasks, rules, and challenges of the National Robotics League. We explain how the robot scores, aligns, lifts, stacks, and navigates under pressure.

WHY WE NEEDED CROWDFUNDING

Even after securing major donations from RoutedAI and Sharon Automation, we still required additional funds for travel, registration, components, fabrication, and documentation materials. Instead of limiting our vision due to budget constraints, we decided to take an independent, student-driven approach: crowdfunding.

HOW WE RAN THE CAMPAIGN

To reach as many people as possible, our team created a simple, accessible, and fully student-built fundraising website.

On this website, we included:

- A clear explanation of our NRL project
- Our mission to make robotics accessible to all
- Photos of our progress
- A secure payment QR code for instant contributions
- A live update section showing how close we were to our funding goal

We then shared this link—along with a heartfelt and persuasive message written by our team—across WhatsApp, email, social media, and personal networks.

We reached out to friends, family, neighbors, teachers, alumni, and anyone who believed in supporting student innovation.

Link to the website: https://nrl-roblocs-008.vercel.app/

MARKETING

MARKETING STRATEGY:

Roblocs' marketing strategy is built around cinematic storytelling that transforms everyday engineering into visually compelling moments. Our philosophy of doing more with less shapes both our robotics and our content: minimal resources, clean visuals, and sequences that highlight sustainability in action: precise decisions, reused materials, redesigned parts, and thoughtfully optimized mechanisms.

Our reels follow a signature style: smooth lighting, close-up textures of components, gentle ambient sounds of tools. The storytelling is subtle but emotionally strong. Instead of drama, we highlight craftsmanship, teamwork, and intelligent rebuilding.

Engagement is maximized through interactive formats — design polls, "choose the better mechanism," rebuild walkthroughs, and Q&A snippets that invite viewers into our process. Community events and outreach content extend this engagement into the real world, creating a loop where online curiosity becomes offline participation.

The strategy positions Roblocs not just as a team but as a creative, sustainable engineering studio, where every frame teaches something, every improvement is traceable, and every viewer feels included in the journey. We don't market noise.

We market thoughtful creation.

SOCIAL MEDIA STRATEGY

Our social media strategy focuses on creating the best robotics content by showcasing every stage of our engineering journey. We highlight CAD design, mechanical assembly, coding, debugging, and testing to present an honest, detailed view of how our robot is built. Along with performance clips and improvement updates, we include short educational posts that explain key concepts and decisions behind our systems.

We maintain a consistent visual identity clean shots, clear captions, and a technical tone to make our content both professional and engaging. By combining engineering depth with high quality storytelling, we create a social presence that stands out in robotics competitions, attracts sponsors, and inspires other teams.

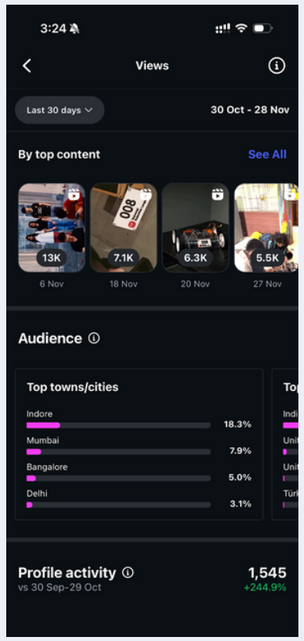
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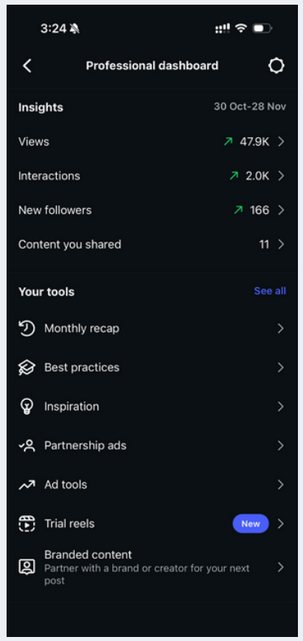
Over the last 30 days, our robotics team's social media performance has shown strong and consistent growth. We recorded 47,866 total views, with 69.4% coming from non-followers, proving that our content is effectively reaching new audiences beyond our existing base. Our content reached 10,593 unique accounts, marking a 205.6% increase from the previous month.

Our top-performing reels received 13K,12k,7.1K, 6.3K, and 5.5K views, showing stable engagement across multiple posts. Interactions totalled 2,000, and we gained 166 new followers during this period.

Audience distribution shows our strongest reach in Indore (18.3%), followed by Mumbai (7.9%), Bangalore (5.0%), and Delhi (3.1%).

Overall, the analytics reflect strong visibility, high non-follower discovery, and engaging robotics content that is consistently performing above average.







OUR MARKETING SPIKE

We focused on showing real engineering, not just polished robot clips. we documented the full build journey, CAD modeling, wiring, debugging, sensor failures, motor burnouts, and fixes in real time. We also used cinematic shots, motion-controlled footage, and clean editing to give our robotics content a professional, high-impact look. Alongside this, we posted short technical explainers that made our page educational, not just promotional. This combination of transparency, storytelling, and technical depth is what made our content stand out and drove our marketing spike.

MARKETING

FUNDRAISING & SPONSORSHIP

Overview
Building a competitive NRL robot requires not only strong engineering skill, but also the ability to secure resources, communicate value, and build meaningful relationships with organizations that believe in our mission. With this in mind, our team reached out to companies across India that support technology education and youth development.
Through this outreach, we were proud to receive major sponsorships from RoutedAI and Sharon Automation. To support the remaining expenses, we launched a crowdfunding initiative—designing a website, writing a personal message, and creating a QR-based payment system. We shared it widely with anyone who might believe in our vision, and every contribution helped bring our robot one step closer to reality.

ROUTEDAI – EMPOWERING INDIA'S YOUTH THROUGH AI

WHO THEY ARE

RoutedAI is an emerging AI startup in India focused on guiding students and young professionals toward meaningful, future-focused careers. Using machine learning, data-driven insights, and personalized pathways, they aim to help the youth of India make informed decisions about their careers.

WHY THEY SUPPORTED US

RoutedAI resonated strongly with our mission to bring robotics education to students across different backgrounds—especially those who lack access to such opportunities.
They saw our NRL initiative as aligned with their core belief: India's youth deserve the right direction, the right technology, and the right opportunities.

THEIR CONTRIBUTION

- Major financial sponsorship supporting our competition needs
- Visibility support, helping us expand our outreach and credibility
- Mission alignment, strengthening our purpose of inspiring young learners through robotics

RoutedAI's donation directly enabled our participation and allowed us to focus on building, refining, and presenting our project with confidence.

SHARON AUTOMATION – SUPPORTING INDIA'S ROBOTICS FUTURE

WHO THEY ARE

Shavron Automations is a robotics and automation company providing advanced solutions across Indian industries. Their work spans manufacturing automation, precision robotics, and hardware innovation.

WHY THEY SUPPORTED US

They believe in encouraging young innovators and supporting school-level robotics initiatives that build India's future engineering talent. Our team's professionalism, documentation quality, and commitment to spreading robotics education aligned with their corporate values.

THEIR CONTRIBUTION

- Financial support as one of our key donors
- Brand endorsement, adding industrial credibility to our project
- Encouragement of youth-driven robotics innovation

While they did not assist in building or designing our robot, their sponsorship played a vital role in enabling our participation and helping us acquire essential resources.

IMPACT OF SPONSORSHIP ON OUR JOURNEY

Even without technical involvement, the support of RoutedAI and Sharon Automation made a tremendous difference. Their contributions helped us:

- Cover competition costs
- Purchase materials, tools, and basic components
- Purchase team jerseys, flags, stickers etc.
- Represent youth robotics with higher confidence and professionalism

These partnerships reflect a shared vision: to make robotics accessible, exciting, and transformative for students across India.

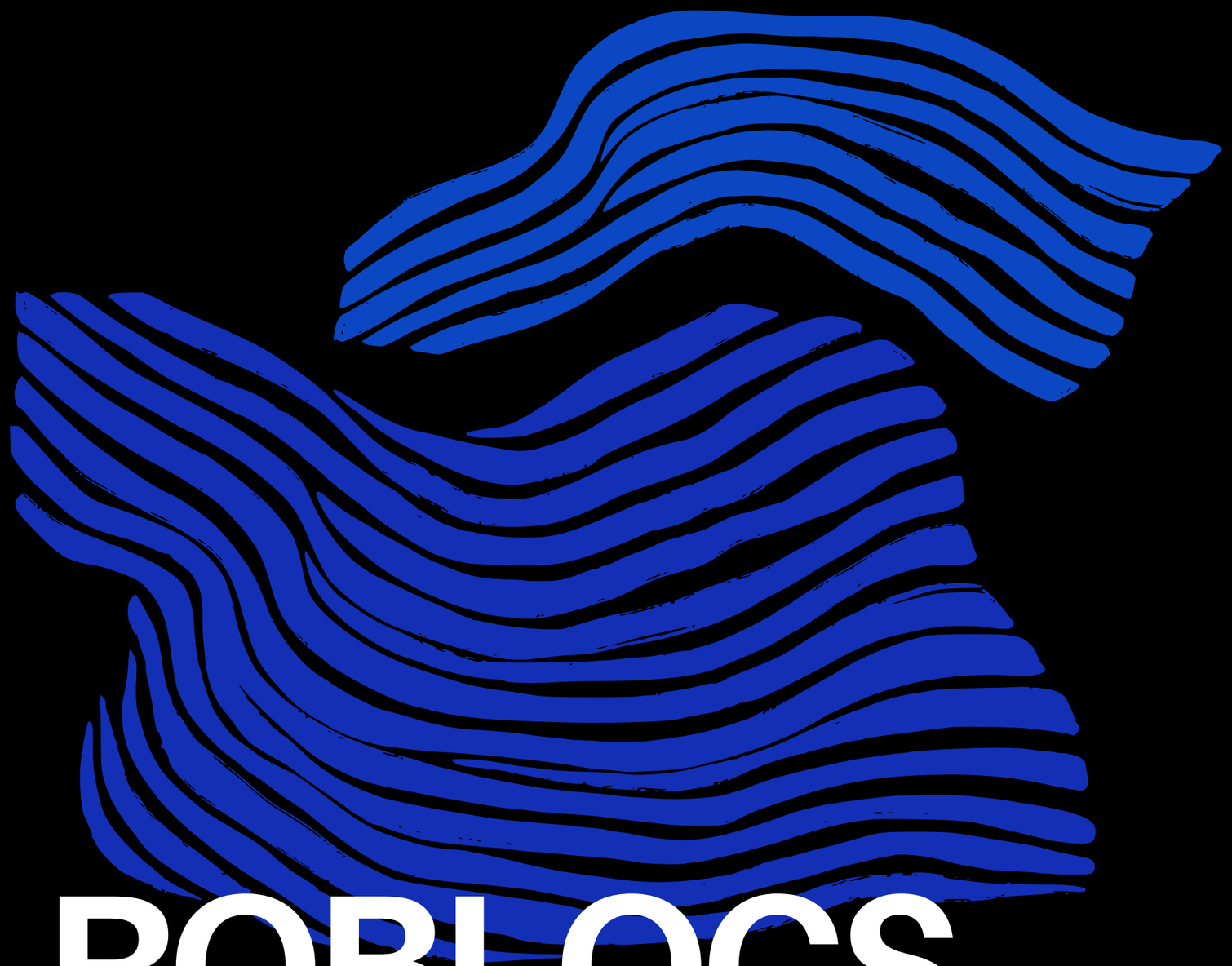
Routed AI

shavRon®

ARTIFICIAL INTELLIGENCE



PORTFOLIO



ROBLOCS